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Late Mesozoic and Cenozoic movements of the Italian Peninsula: Further paleomagnetic data from the Umbrian sequence

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ABSTRACT

About 210 samples of rocks of late Early Jurassic to early Tertiary age from 37 sites in the Umbrian carbonate sequence in the Northern Apennines were investigated paleomagnetically. Special attention was paid to the Scaglia Formation of Cenomanian to middle Eocene age, and three well-correlated sections were studied in detail. Accurate relative ages were determined from the well-preserved planktonic foraminiferal faunas.

Paleomagnetic directions of Toarcian-Aalenian to late Aptian sites group fairly well, which implies the absence of major rotational and latitudinal movements of the sampling area during that interval. An only minor phase of movement at about Tithonian time resulted in a counterclockwise rotation over approximately 10°. Late Aptian to late Cenomanian directions indicate a southward movement over 10° of latitude; this was followed by a 30° counterclockwise rotation during Turonian, Coniacian, and Santonian time, and in addition, a northward movement over 10° of latitude during Santonian and Campanian time. Maastrichtian to early Eocene data group reasonably well but also deviate from the present local field direction; this indicates a further 25° counterclockwise rotation and an additional northward movement over more than 10° post–early Eocene time.

We suggest that the data presented here from the Northern Apennines are representative for the general pattern of movements of the Italian Peninsula.

These paleomagnetically deduced movements conform during at least late Early Jurassic to early Tertiary time with the movements of the African plate as established from sea-floor spreading and paleomagnetic data. At the same time, however, the paleomagnetic results indicate that this coupled motion of the Italian Peninsula and the African continent did not persist until the present: a post–early Eocene 25° counterclockwise rotation of the Italian Peninsula must have occurred relative to Africa.

Our data are compared with available paleomagnetic data from the western Mediterranean area. Their relation to the structural evolution of the areas concerned is discussed.

INTRODUCTION

The quantity of paleomagnetic data from the Mediterranean area is rapidly increasing because many groups have focused their research on this complex area. Paleomagnetic investigations are relatively important, since magnetic anomaly patterns in the Mediterranean are not well developed and give little or no information about megatectonic history. Recently proposed models for the structural evolution of the Mediterranean, like those of McKenzie (1970) and Dewey and others (1973), imply a complex pattern of plates and microplates moving relative to each other. The postulated movements of these plates can be tested by establishing standard paleomagnetic directional sequences in autochthonous elements on the plates.

Conclusions on the megatectonic history of the Mediterranean have been drawn mainly from well-established magnetic anomaly patterns in the North Atlantic (Smith, 1971; Phillips and Forsyth, 1972; Pitman and Talwani, 1972; Dewey and others, 1973). These analyses of the movements of the African plate and Europe, based on the successive opening phases of the Atlantic, give little information on the movements of minor plates in the enclosed Mediterranean area. Delineation of these minor plates and determination of their relative movements are problems yet to be solved. Paleomagnetic data are crucial in this respect.

Zijderveld and Van der Voo (1973) postulated that major parts of Mediterranean Europe formed part of the African plate during its post-Hercynian history. Their hypothesis was mainly based on the following observations: (1) The then available pole positions from the Iberian Peninsula and the Southern Alps conform with the apparent polar wander path of Africa. (2) Well-established paleomagnetic data from some major units of Mediterranean Europe indicate similar rotations of comparable magnitude. (3) A synchronism in rotation of the Iberian Peninsula (between Late Jurassic and Eocene time) and Southern Alps (between Middle Triassic and Eocene time) with the rotation of the African plate (between Early Cretaceous and Eocene time) seems probable. Schult (1973) put forward a similar conclusion for Sicily. However, evidence obtained by Soffel (1972, 1974, 1975b) for a late Eocene to early Oligocene rotation, as shown in volcanic rocks from the northern Po Plain area, and Sardinia's Oligocene to early Miocene rotation (De Jong and others, 1973; Coulon and others, 1974) suggest that Zijderveld and Van der Voo's basically simple hypothesis needs further testing and elaboration.

GEOLOGIC SETTING, SAMPLING, AND DATA ANALYSIS

The paleomagnetic investigations were focused on the Mesozoic Umbrian carbonate sequence. In contrast to the adjacent Liguride and Tuscan sequences, developed largely in nappe structures (allochthonous), the Umbrian sequence is generally supposed to be autochthonous.

Outcrops of the Umbrian carbonate sequence are located in the southeastern part of the Northern Apennines and form a regularly curved belt about 200 km long and 60 km wide at its greatest width. This belt (Fig. 1) stretches southward from the Cagli area (lat 43°33'N, long 12°41'E), near the classical stratigraphic section of the Bottaccione gorge at Gubbio, near Rieti (lat 42°25'N, long

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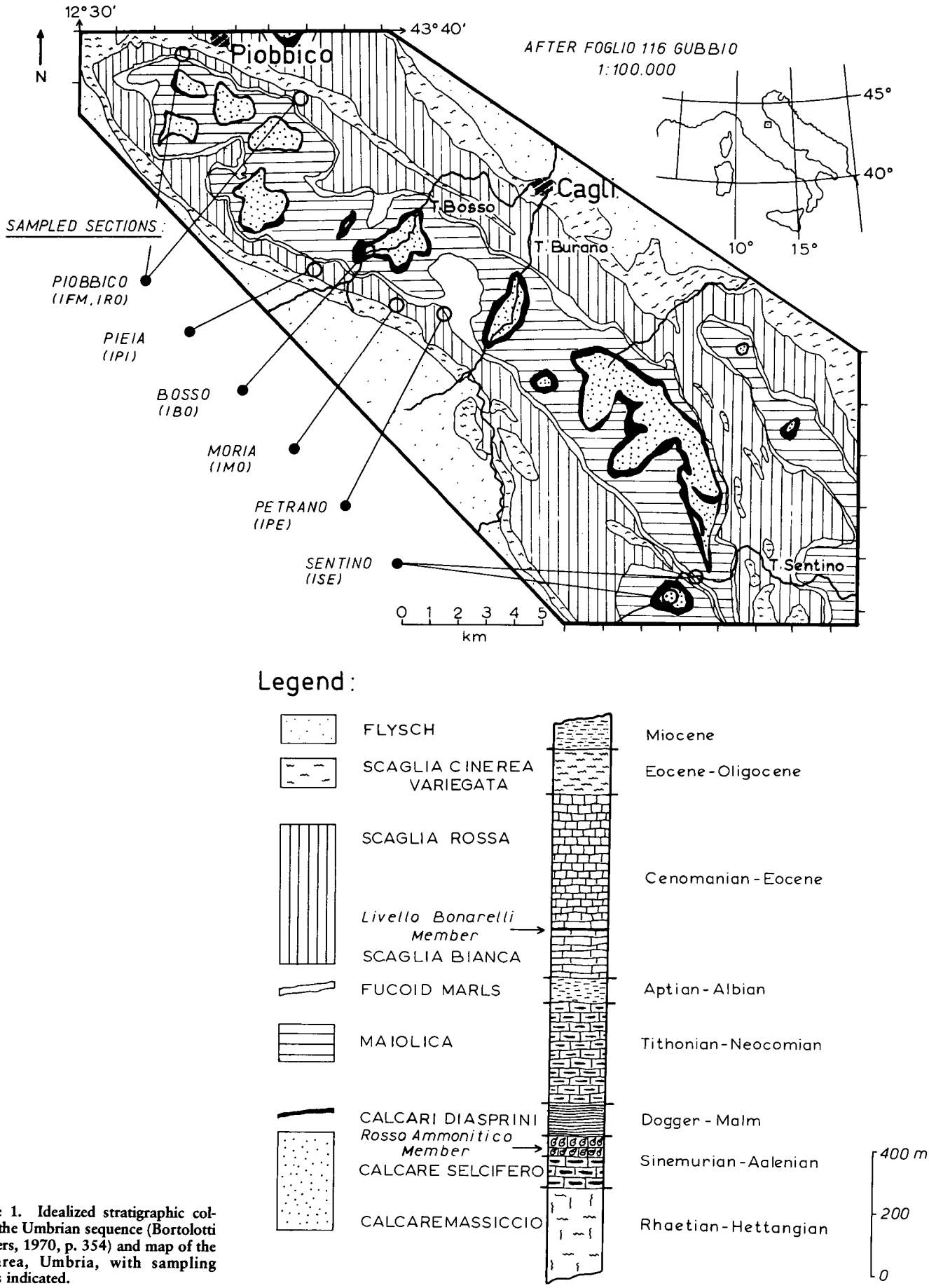


Figure 1. Idealized stratigraphic column of the Umbrian sequence (Bortolotti and others, 1970, p. 354) and map of the Cagli area, Umbria, with sampling localities indicated.

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12°52'E); the belt is truncated to the southeast by the Anzio-Ancona line.

We mainly directed our attention to the Cagli area, where the Umbrian sequence occurs in a wide northwest-trending anticline with a slight northwest plunge. The moderately dipping southwestern limb proved to be particularly attractive for paleomagnetic sampling. The steeply dipping, and locally overturned, northeastern limb is of complex structure; therefore, we restricted the sampling here to an application of the fold test only.

The Umbrian sequence is generally considered to be of miogeosynclinal nature and its lithology, as represented in the idealized stratigraphic colum (Fig. 1), was described in detail by Bortolotti and others, (1970).

In the sampling area, the outcropping rocks grade from a basal (Upper Triassic to Lower Cretaceous) sequence of limestones with some marly and siliceous intercalations (that is, the Calcare Massiccio, Calcare Selcifero [including the Rosso Ammonitico Member], Calcari Diasprini, and Maiolica) into the middle Cretaceous Fucoid Marls, which form the base of huge escarpments. The overlying limestones and marly limestones of the Scaglia Formation, which is informally subdivided into a lower "Scaglia Bianca" and an upper Scaglia Rossa unit, are well exposed in these escarpments. The lithologic boundaries between the units of the Scaglia Formation are quite irregular and probably not isochronous. A remarkable lithologic marker in the Scaglia Bianca is the bituminous Livello Bonarelli Member, a black shale with radiolaria and as much as 20% organic matter. The Scaglia Rossa, the youngest unit sampled for this study, ranges into the middle Eocene part of the section.

Preliminary sampling (10 sites, 43 samples) in the summer of 1973 appeared to be successful (Klootwijk and VandenBerg, 1975), and further detailed sampling was carried out in the summer of 1974 (37 sites, 210 samples). Especially in the Scaglia Formation, the sites were located in continuous sections (rather than being geographically distributed) in order to detail the sequential order of paleomagnetic direction changes. Red Scaglia limestones were sampled in three sections of about 10 sites each in fresh road cuts in the southern limb of the anticlinal structure — namely along the Pianello-Pieia road, the Moria–Monte Petrano road and the Monte Petrano–Ponte d'Azzo road (Fig. 1). In addition, shorter sections and some separate sites were sampled in the Rosso Ammonitico, in a siliceous red limestone bed in the Calcari Diasprini, in the basal Maiolica beds, in the Fucoid Marls, and, again, in the Scaglia Formation. The additional samples from the Fucoid Marls and Scaglia Formation were collected from the plunging nose and the northern limb of the anticline for an accurate application of the fold test. At each site, and in more detail throughout the sections, special samples were collected for micropaleontologic studies. The rich and well-preserved planktonic foraminiferal faunas of the Umbrian sequence were promising for the construction of a rather continuous and well-dated paleontologic record of successive movements of the Northern Apennines.

MAGNETIC PROPERTIES

In the laboratory, core specimens (2.2-cm height, 2.5-cm diameter) were drilled from the hand samples. All measurements were done at the Paleomagnetic Laboratory of the State University of Utrecht by means of the Czechosolvakian JR-3 spinner (Jelinek, 1966).

The specimens were subjected to standard alternating field (AF) demagnetization and, for a few sites, to thermal demagnetization as well. At least one pilot specimen from each site was progressively demagnetized in 10 to 15 steps up to the maximum obtainable peak value of 3,100 Oe or up to 700 °C. Following these pilot demagnetizations, at least five other specimens from each site were partially progressively demagnetized in 6 to 10 steps in the appropriate alternating field or thermal range.

Through analysis of orthogonal projections (Zijderveld, 1967, 1975), showing sequential changes of the magnetization vector upon demagnetization, the characteristic and secondary magnetization components could be separated and their directions determined. For computation of mean directions, Fisherian statistics were used.

The intensity of initial remanent magnetization of the samples was very weak, generally ranging from 10^{-5} to 10^{-7} gauss. Initial specimen directions from all sites were distributed mainly in the lower hemisphere (Fig. 5A), with a clear tendency for concentration toward the northern quadrants. This is largely due to an appreciable secondary component related to the present local field ($D = 359°$, $I = +59°$).

In total, three vectorial components could be distinguished in the progressive demagnetizations. A soft secondary magnetization component of large intensity and aligned according to the direction

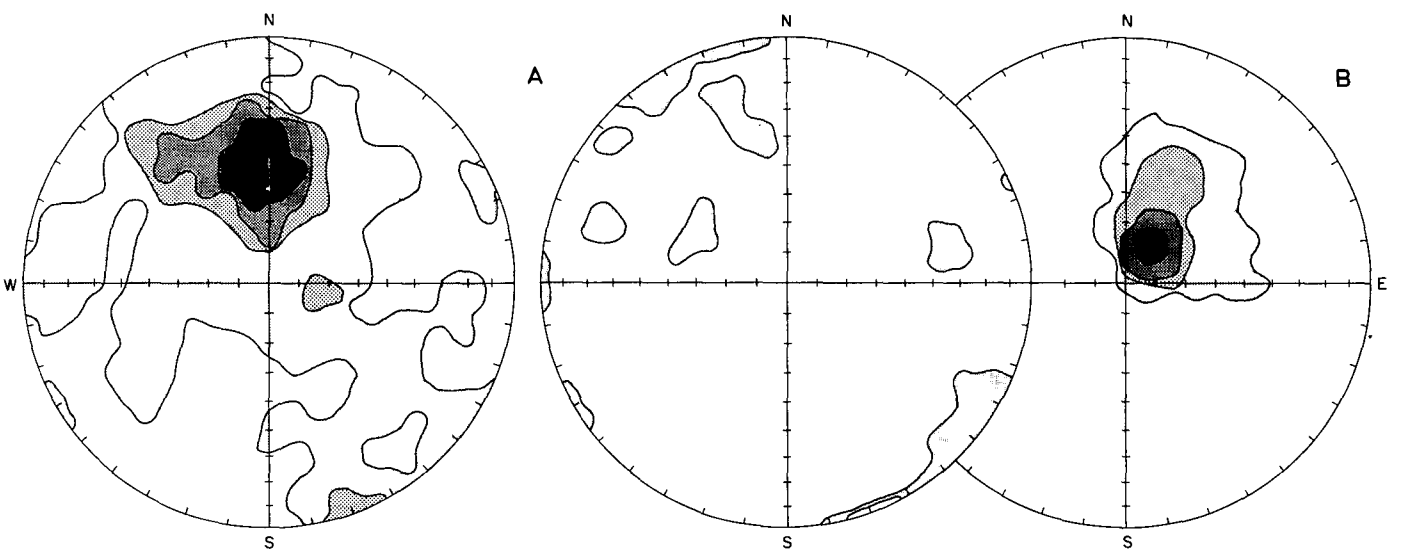
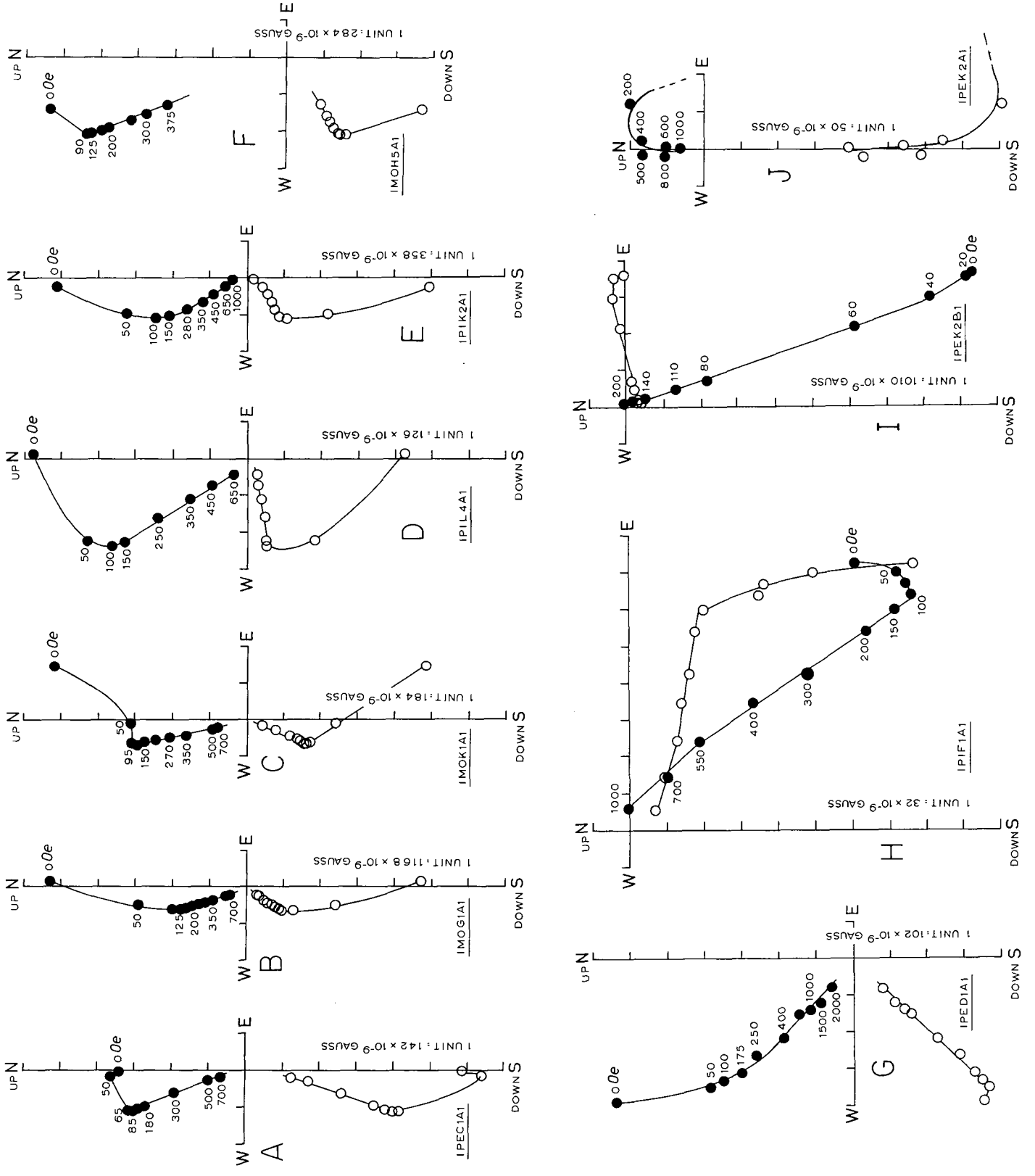


Figure 2. Density distribution of remanent magnetization directions in equal-area projection, top view. The asterisk (white on black) denotes the present local field in the sampling area. (A) Initial remanent magnetization directions, lower-hemisphere (left) and upper-hemisphere projections (right).(B) Soft secondary magnetization components in lower-hemisphere projection.

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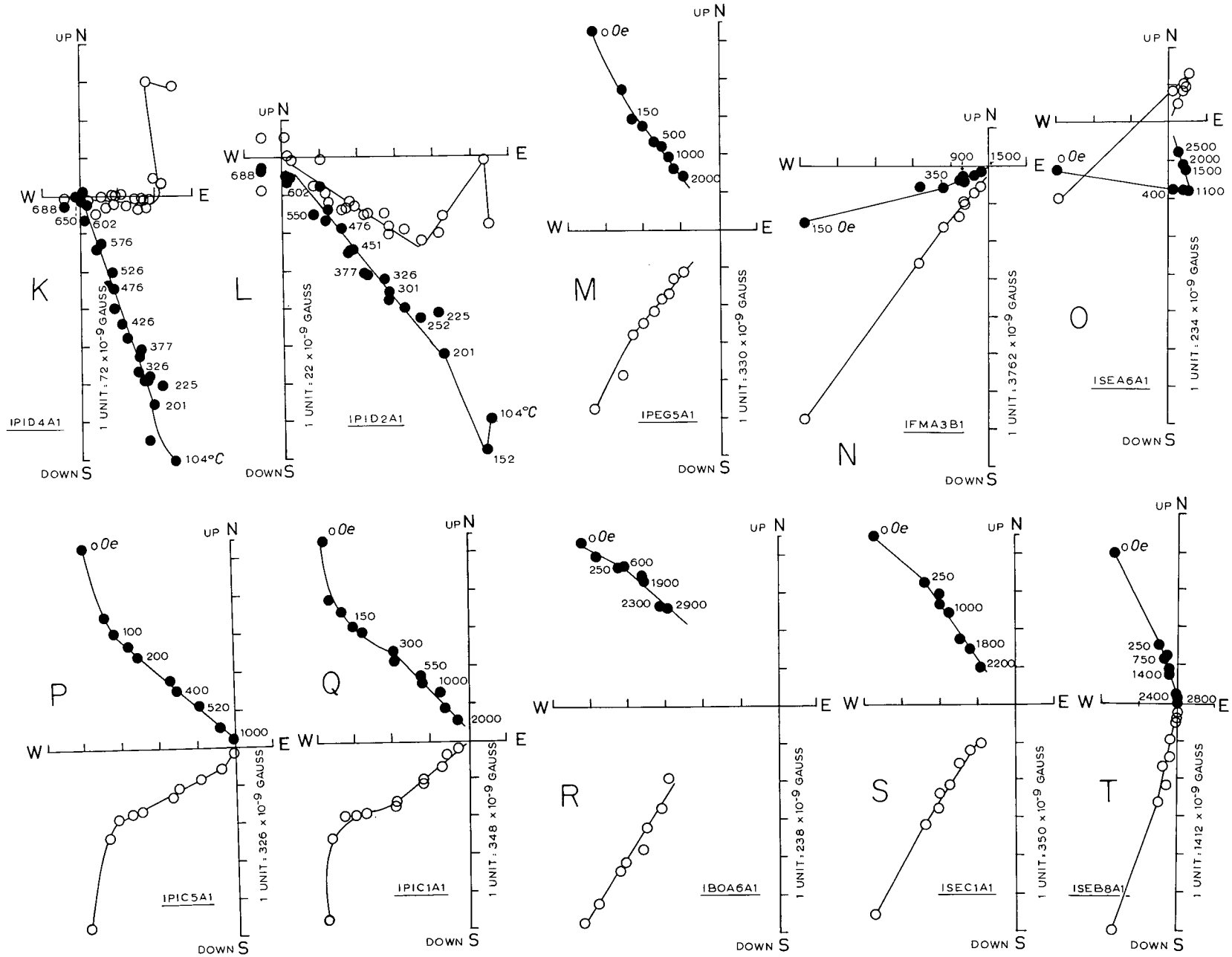


Figure 3. Orthogonal projections diagrams of the vector path of the total remanent magnetization vector during AF cleaning (A through J, M through T) or thermal cleaning (K, L). Circles are projections on the vertical east-west plane; dots are projections on the horizontal plane. Numbers denote peak values of the applied alternating fields (in oersteds) or values of the applied temperatures (in degrees Celsius). Underlined specimen code given in each diagram. Diagram J represents projections on an enlarged scale of diagram I, for the final demagnetization steps only. Note that 1 gauss $\equiv 10^3$ A/m; 1 Oe $\equiv 10^{-4}$ T (tesla).

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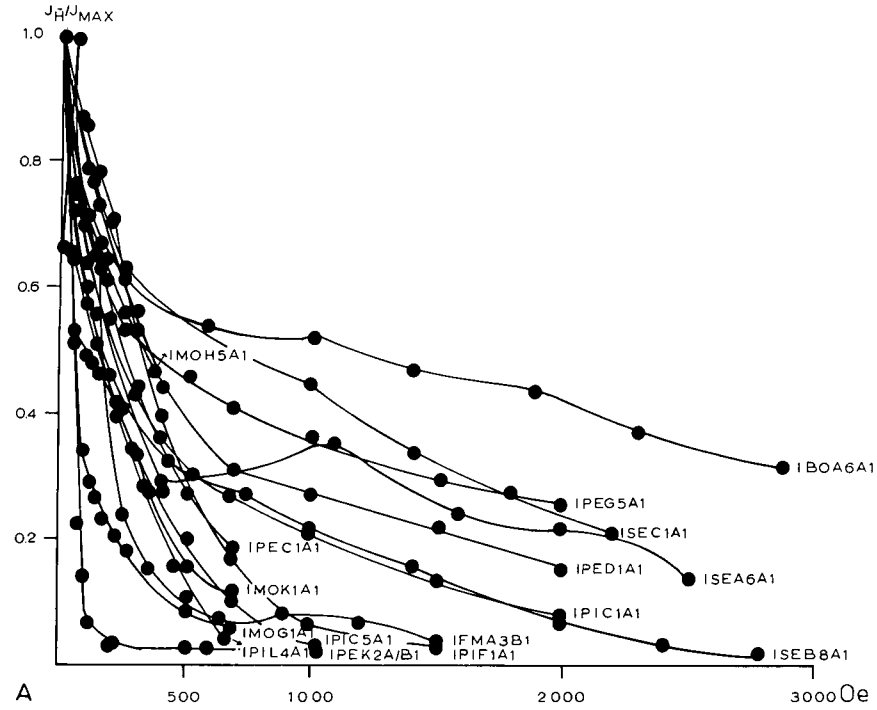
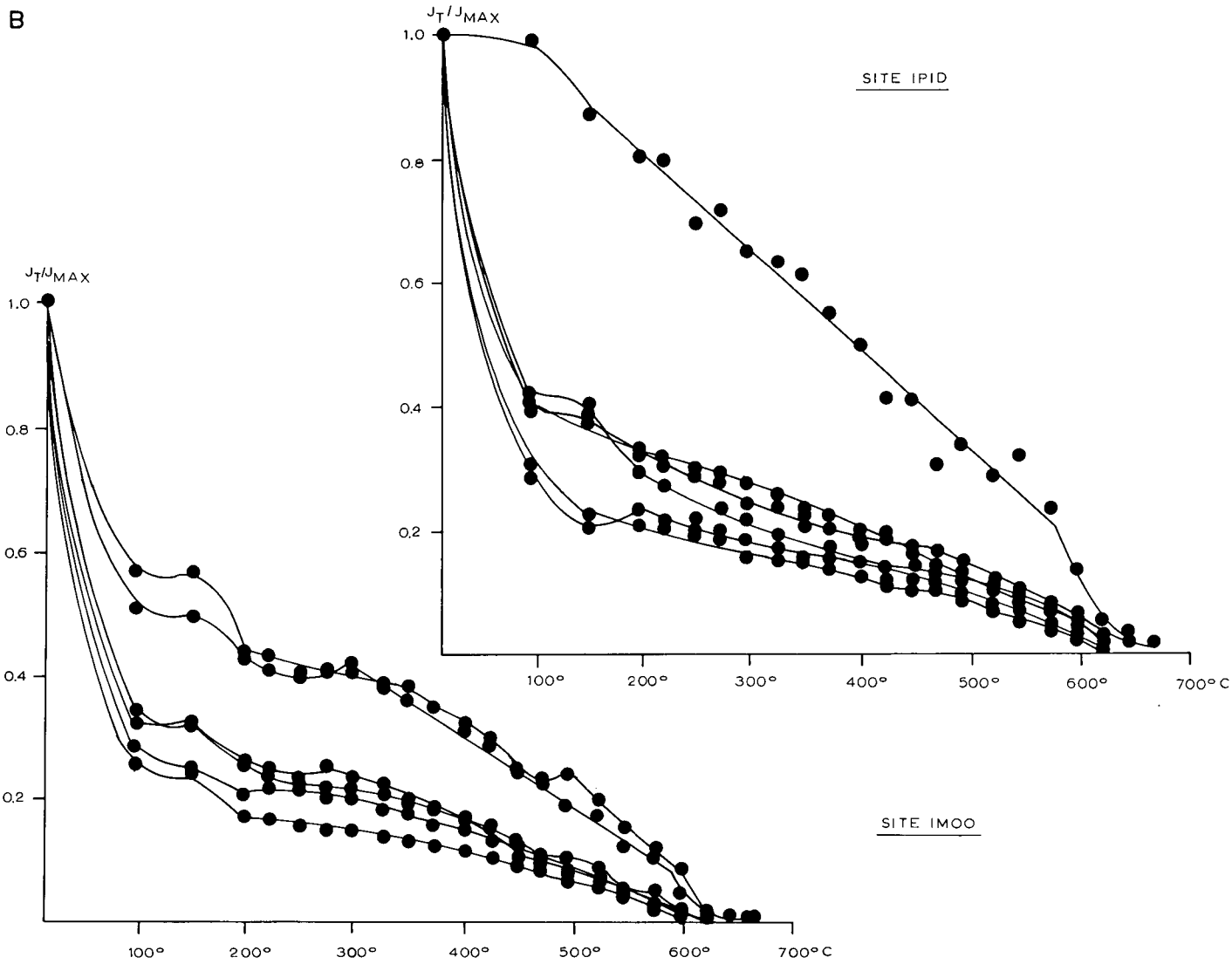


Figure 4. Normalized curves, showing the decrease of total remanent magnetization during AF cleaning (A) and thermal cleaning (B). J_H/J_{max} and J_T/J_{max} are normalized intensities of the natural remanent magnetization during alternating field and thermal cleaning, respectively.



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of the present local field or, in case of the Scaglia samples with a somewhat more northeasterly declination (Fig. 2B), was generally removed upon AF demagnetization at 100-Oe to sometimes 200-Oe peak value. From 100 to 200 Oe onward up to 500 to 700 Oe for the Scaglia samples and up to much higher peak values for the samples of other formations, the characteristic magnetization component was eliminated. In the Scaglia samples, another hard secondary magnetization component was found, also aligned to the present-day local field. This hard secondary magnetization could be generally removed between 500 and 3,000-Oe peak value. This removal occurred sometimes separately from, sometimes simultaneously with, a remaining part of the characteristic magnetization; thus the directions of these very hard secondary magnetizations were sometimes difficult to determine accurately.

Scaglia Rossa and Bianca

The characteristic remanent magnetizations of the Scaglia samples are mainly of normal polarity, but reversed polarity directions were noted in its upper part (sites IMOO, IPID, IMOM, IPIF, IPEK, ISEA, IBOC, Table 1). In all Scaglia samples (Figs. 3A through 3H) a mainly northeast to north directed, steeply downward dipping secondary component of appreciable intensity — that is, 25% to 75% of the initial intensity (Fig. 3, 4A) — was removed upon AF demagnetization at about 100- to sometimes 200-Oe peak value. This soft secondary component, resembling the present local field direction, was present in both the samples of normal polarity (Figs. 3A through 3G) and of reversed polarity (Fig. 3H). In Figure 3I and 3J, we show a special case in which a very small soft secondary component is present in a reversed polarity sample. Well-concentrated characteristic directions were removed in general between 100-Oe and 700-Oe peak values (Figs. 3A through 3D, 3H), with a slight part of the characteristic component residing in domains with higher coercive forces (Figs. 3E, 3G), Sometimes a third magnetization component was removed upon demagnetization above 700 Oe. This component was identified before (Klootwijk and VandenBerg, 1975) as another "present local field" component residing in domains with higher coercive forces. Although good AF demagnetization results were obtained for a number of sites (Figs. 3H through 3J), thermal demagnetization was the preferred technique for site IPID as shown in Figures 3K and 3L. Upon thermal

TABLE 1. NATURAL REMANENT MAGNETIZATION

Site	No. of samples	Initial intensity ($\times 10^{-8}$ gauss)	Site-mean direction* (°)	(°)	α_{95} (°)	Age	Mean direction* (°)	(°)	α_{95} (°)	Pole position lat (°N)	long (°W)	δp (°)	δm (°)
IMOO	5	27–113	149.2	−50.8	5.1	Late Paleocene to late Eocene	150.1	−49.7	3.0	63.0	96.0	2.7	4.0
IPID	11	15–159	150.5	−49.2	4.1								
IMOM	6	218–378	148.8	−44.9	3.8	Late Campanian to early Maastrichtian	329.7	+48.2	3.7	61.9	98.9	2.8	4.2
IPEL	5	387–996	338.5	+52.8	5.5								
IPIF	5	27–51	143.2	−46.9	4.6								
IMOL	5	119–196	326.2	+55.1	5.1	Early to late Campanian	328.1	+54.7	3.6	64.2	85.2	5.5	5.1
IPIG	5	141–701	330.1	+54.3	6.7								
IPEK	5	813–1,836	141.1	−40.6	6.7	Early Campanian	328.8	+43.3	6.1	58.7	102.9	4.7	7.6
.IPEH	5	82–160	337.2	+45.3	8.8								
IMOK	6	113–153	334.8	+40.5	3.9	Late Santonian	328.5	+43.3	4.2	58.4	102.7	3.2	5.3
IPEA	6	86–142	326.5	+44.1	11.1								
IPIH	6	52–145	323.8	+44.6	7.5								
IPEB	6	57–112	321.3	+35.3	6.5	Late Coniacian to early Santonian	320.3	+36.6	2.7	49.7	99.5	1.8	3.1
IMOH	5	30–211	322.2	+38.1	3.6								
IPIL	6	78–156	317.7	+36.7	4.8								
IPEC	4	96–180	310.9	+36.1	12.5	Coniacian	307.1	+34.4	6.1	39.5	89.1	4.1	7.0
IPED	5	77–120	304.2	+32.9	8.0								
IMOG	5	165–2,829	310.8	+35.4	10.9	Early Turonian	310.9	+36.9	5.4	43.3	90.6	3.7	6.3
IPIK	5	188–1,484	311.0	+38.4	6.5								
IPEF	5	547–21,471	300.7	+37.4	3.3	Late Cenomanian	300.5	+39.2	4.3	36.9	80.9	3.1	5.2
IMOF	5	361–2,044	300.4	+40.9	9.5								
IFMA	5	410–3,880	291.7	+44.1	12.2	Late Aptian	300.2	+47.0	4.4	40.3	74.5	3.7	5.7
ISEA	6	102–112	127.8	−47.6	4.7								
IFMB	6	410–3,880	307.2	+51.6	5.8								
IPEG	10	104–259	296.3	+44.8	9.9								
IPIA	5	184–557	304.7	+53.1	10.7	Early Neocomian	298.6	+47.6	7.0	39.4	72.9	5.9	9.1
IPIC	5	241–279	293.7	+41.9	7.7								
IBOA	6	167–358	289.3	+48.4	16.2	Kimmeridgian to Tithonian	289.9	+50.0	8.0	34.4	65.4	7.1	10.7
IBOB	7	66–595	290.4	+51.3	9.6								
IBOC	4	123–325	109.6	−41.9	29.1								
ISEB	8	65–2,032	290.2	+51.2	9.3	Toarcian to Aalenian	294.7	+50.6	6.4	38.0	67.7	5.8	8.6
ISEC	5	43–53	301.4	+49.2	9.5								
IROC	6	65–131	321.4	+34.4	8.5								
IROD	5	76–132	289.3	+30.1	9.8								

Note: α_{95} = radius of the circle of confidence on the 95% level.
(δp, δm) = the oval of confidence on the 95% level about the pole position (δp = error in paleolatitude; δm = error in paleolongitude).
* First column is azimuth; second column is inclination. Positive inclinations represent downward-pointing directions.

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demagnetization of the samples from this site, the present local field component was removed around 100 to 150 °C. The direction of the characteristic magnetization component could be determined accurately upon progressive demagnetization between 250 and almost 600 °C (Fig. 3L) or higher (Fig. 3K). After thermal demagnetization up to the Curie temperature of magnetitie (about 580 °C), the small remaining magnetization (Fig. 4B), with a Curie temperature of about 650 °C, indicated hematite to be present as well. As shown in Figure 3K, this small hematite magnetization component has a direction similar to the characteristic magnetization residing in the magnetite. A similar observation was noted also by Lowrie and Alvarez (1975).

Three sites in the upper part of the Scaglia Rossa in the northwest-plunging nose of the anticlinal structure yielded dispersed and uninterpretable results. These are probably due to the effect of drag folds, which are at places extensively developed in the Scaglia Formation (Fazzini, 1973).

The fold test, applied to samples from both the gently dipping southern limb and the subvertical northern limb of the anticlinal structure (Fig. 5), clearly indicates the characteristic magnetization component to be of pretectonic, that is, pre-Pliocene, origin. As concluded earlier (Klootwijk and VandenBerg, 1975), we interpret this characteristic magnetization component as the primary magnetization. A similar conclusion was reached by Lowrie and Alvarez (1975) from detailed analysis of the magnetic mineralogy of Scaglia Rossa samples.

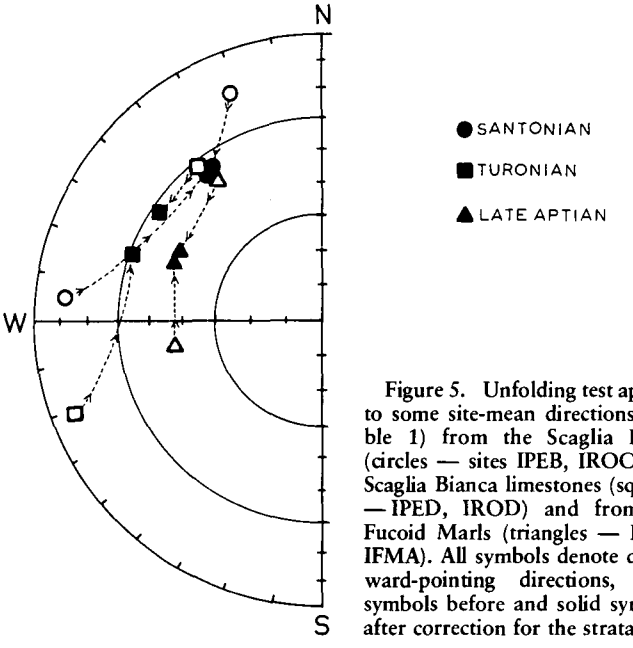


Figure 5. Unfolding test applied to some site-mean directions (Table 1) from the Scaglia Rossa (circles — sites IPEB, IROC) and Scaglia Bianca limestones (squares — IPED, IROD) and from the Fucoid Marls (triangles — IPEG, IFMA). All symbols denote downward-pointing directions, open symbols before and solid symbols after correction for the strata dip.

Fucoid Marls

In the Fucoid Marls (Figs. 3M, 3N) a present local field component of about 40% of the initial intensity was removed at about 150 Oe. Well-grouped characteristic magnetization directions, mainly of normal polarity, were removed above 250 Oe. Apart from the upper Scaglia Rossa samples, reversed polarity directions were observed only in the samples from site ISEA (Fig. 3O), taken in the basal part of the Fucoid Marls in the Sentino gorge. In these samples a secondary magnetization component of great intensity, that is, 70% of initial intensity, with downward inclination and of uninterpretable westward declination, was removed at about 1,100-Oe peak value. From 1,100 Oe onward, well-determined characteristic directions of reversed polarity were removed. Samples from both the gently dipping southwestern limb and from the gently dipping nose of the anticlinal structure convincingly show this characteristic magnetization to be of pretectonic origin (Fig. 5).

Basal Miaolica Beds

In the basal Maiolica beds, a present local field component of about 25% to 50% of the initial intensity was removed at about 150-Oe peak value, with the characteristic magnetization component being removed between 150- to 300-Oe and 2,000-Oe peak value (Figs. 3P, 3Q, 4A). Some samples also seemed to show a second present local field component, although simultaneous removal with the remaining part of the characteristic magnetization prevented a clear interpretation.

Calcari Diasprini

In the Calcari Diasprini samples, a northwest-trending, downward-directed secondary magnetization component of about 30% of initial intensity was removed at 250-Oe peak value (Figs. 3R, 4A). The characteristic magnetization appeared to be of very hard character and was only partly removed at the obtainable peak value of AF demagnetization. Some samples had to be rejected since they contained only secondary magnetization components.

Ammonitico Rosso

In the Ammonitico Rosso samples a northwest-trending, downward-directed secondary magnetization component of about 35% to 50% of initial intensity and probably of present local field origin could be removed at 250 Oe (Figs. 3S, 3T, 4A). Upon progressive demagnetization above 250 Oe in both marl and limestone samples, well-determined characteristic directions were obtained.

PALEONTOLOGIC DATA

Micritic carbonates of marine pelagic origin prevail throughout the Umbrian sequence, and marine pelagic fossils are abundantly present in the carbonates as well as in the few intercalations of deposits of more terrigenous clastic character. The Umbrian sequence, and the Scaglia Formation in particular, is a classic deposit in Cretaceous planktonic biostratigraphy. In the Gubbio area, which is near the region we sampled around Cagli, the Scaglia Formation is developed in its most complete and undisturbed pelagic facies, comparable to that of recent planktonic foraminiferal oozes. It was in the Bottaccione section near Gubbio that the first subdivision of the Upper Cretaceous Series by means of planktonic foraminifera was made by Renz (1936). His four biozones of the Upper Cretaceous Series, based on thin sections only, have been shown to be reliable throughout the Tethyan realm.

Study of isolated planktonic foraminifera established many more zones, all of which can be recognized in the Umbrian Scaglia Formation (Luterbacher and Premoli Silva, 1962, 1964). This continuous fossil record of the sampled Scaglia sections allowed accurate dating of the paleomagnetic observations. Thin sections were prepared from the same hand samples from which core specimens for paleomagnetic analysis were taken, in order to make errors as small as possible. The washing residues of some additional samples were studied in order to check the determinations based on thin sections. In order to make our results reproducible, the species (Figs. 6A, 6B, 6C) were taken in the broadest taxonomic sense. The relative age determinations were based on the zonations of Moullade (1966) for the Albian and Aptian rocks, and those of Postuma (1971) for the Upper Cretaceous and lower Tertiary rocks.

The results reveal a regular age versus deposition curve (Fig. 7). Only three samples yielding Paleocene and Eocene faunas are in a relatively low stratigraphic position when compared to the roughly linear relationship during the Cretaceous Period. This is possibly due to the suspected hiatus at the Cretaceous-Tertiary boundary (Luterbacher and others, 1964), though a drastic reduction in a sedimentation rate cannot be excluded (Premoli Silva and others, 1974).

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The taxa shown in the distribution charts (Fig. 6) are listed below with short comments. A rather wide species concept had to be applied, as nearly all determinations are based on thin sections. The names given stand for morphologic groups in which several species would probably be recognized if they could have been studied in washing residues.

Hedbergella trocoidea (Gandolfi)

Anomalina lorneiana var *trocoidea* Gandolfi (1942, p. 134, P1. 2, fig. 1a–c).
Abundant in the lower part of the Fucoid Marls. Determination was confirmed by washing residues.

Globigerinelloides algerianus Cushman and Ten Dam

Globigerinelloides algerianus Cushman and Ten Dam (1948, p. 43, P1. 8, figs. 4–6).
This species was found both in washed samples and in thin sections. Some individuals may belong to the group of *Planomalina cheniourensis* Sigal (Sigal, 1966). We unite all nonkeeled forms in the group of *G. algerianus*.

Planomalina cheniourensis Sigal

Planomalina cheniourensis Sigal (1966, p. 24, Pl. 3, figs. 5, 8, 9, 11).
Only keeled forms are considered to belong to this species.

Rotalipora appenninica (O. Renz)

Globotruncana appenninica O. Renz (1936, p. 14, Fig. 2).
We include all relatively flat, biconvex *Rotalipora* under this name. A washed sample (IPEO) yielded great numbers of typical forms.

Roralipora greenhornensis (Morrow)

Globorotalia greenhornensis Morrow (1934, p. 199, Pl. 31, fig. 1).
This species was found to be associated with *R. cushmani*. We use the specific name in a broad sense; forms resembling *R. brotzeni* (Sigal) may be included.

Rotalipora cushmani (Morrow)

Globorotalia cushmani Morrow (1934, p. 199, Pl. 31, figs.2 a–b, 4a–b).
The strongly inflated chambers on both sides of the test are characteristic.

Praeglobotruncana stephani (Gandolfi) s. l.

Globotruncana stephani Gandolfi (1942 , p. 130, Pl. 3, fig. 4).
On the basis of as few individuals as were present in two or three thin sections, it was impossible to distinguish the main species or subspecies of this group, which therefore also comprises *P. delrioensis* (Plummer) and *P. turbinata* (Reichel).

Praeglobotruncana helvetica (Bolli)

Globotruncana helvetica Bolli (1945, p. 226, Pl. 9, fig. 6).

Globotruncana renzi Gandolfi

Globotruncana renzi Gandolfi (1942, p. 124, Text-Fig. 45, Pl. 3, fig. 1a–c.
In thin sections, this species generally appears as biconvex forms with closely spaced keels. At the younger chambers, the keel may be single or even lacking. From a current study (A.A.H. Wonders, in prep.), we know that forms transitional to both *G. sigali* and *G. coronata* may be found. Together with the latter two species, *G. renzi* determines the general aspect of the extensive zone with large, flat *Globotruncana,* which can be subdivided by means of representatives of the *G. concavata* group.

Globotruncana sigali Reichel

Globotruncana (Globotruncana) sigali Reichel (1950, p. 610, Pl. 16, fig. 7).
See remarks on *G. renzi*.

Globotruncana coronata Bolli

Globotruncana lapparenti coronata Bolli (1945, p. 233, Fig. 1, nos. 21, 22, Pl. 9, figs. 14, 15).
See remarks on *G. renzi*.

Globotruncana primitiva Dalbiez

Globotruncana (Globotruncana) ventricosa primitiva Dalbiez (1955, p. 168, Text-Fig. 6).
It is possible to distinguish the three members of the *G. concavata* group in thin sections. *G. primitiva* is characterized by moderately inflated chambers, a nearly flat spiral side, and a low convex central side of the test. The keels are very close to each other, as in all representatives of the group.

Globotruncana concavata (Brotzen)

Rotalia concavata Brotzen (1934, p. 66, Pl. 3, fig. b).
The typical, highly inflated chambers at the ventral side make this species easly recognizable. See also remarks on *G. primitiva*.

Globotruncana carinata Dalbiez

Globotruncana (Globotruncana) ventricosa carinata Dalbiez (1955, p. 168, Text-Fig. 8).
The chambers are strongly inflated at the ventral side and subangular in outline. Periumbilical ridges are present. See also remarks on *G. primitiva*.

Globotruncana linneiana (d'Orbigny) s.l.

Rosalina linneiana d'Orbigny, (1839, p. 101, Pl. 5, figs. 10–12.
Under this name, we include all biplanar forms with widely spaced keels.

Globottruncana elevata (Brotzen) s.l.

Rotalia elevata Brotzen (1934, p. 66, Pl. 3, fig. c).
This name stands for all ventroconvex single-keeled forms, which may belong to *G. elevata, G. stuartiformis* Dalbiez, or *G. stuarti* (de Lapparent).

Globotruncana fornicata Plummer

Globotruncana fornicata Plummer (1931, p. 198, Pl. 13, figs. 4–6).
The large group of *G. fornicata* has been subdivided into several species. Here, we only distinguish three species: *G. renzi* (biconvex), *G. fornicata* (moderately dorsoconvex), and *G. contusa* (Cushman). *G. contusa* is an extremely high dorsoconvex species and was not found in our matrial.

Globotruncana rosetta (Carsey) s.l.

Globigerina rosetta Carsey (1926, p. 44, Pl. 5, fig. 3).
The taxonomic position of this group is uncertain. In our concept, it comprises forms of variable convexity with a weakly developed second keel, and — often not visible — an extensive and very high tegillar structure.

Globotruncana arca (Cushman)

Pulvinulina arca Cushman (1926, p. 23, Pl. 3, fig. 1).
Distinctly dorsoconvex forms with widely spaced keels are united under this name.

Globotruncana ventricosa (White)

Globotruncana canaliculata var. *ventricosa* White (1928, p. 284, Pl. 38, fig. 5a–c).
This species is easily recognized by its high ventroconvex outline and relatively widely spaced keels.

Heterohelicidae

Representatives of this family were determined only at the generic level *(Heterohelix, Planoglobulina, Pseudotextularia).* They were not used for age determinations.

Globorotalia velascoensis (Cushman)

Pulvinulina velascoensis Cushman (1925, p. 19, Pl. 3, fig. 5a–c).
The angular, strongly ventroconvex outline and heavy periumbilical ridges are characteristic features.

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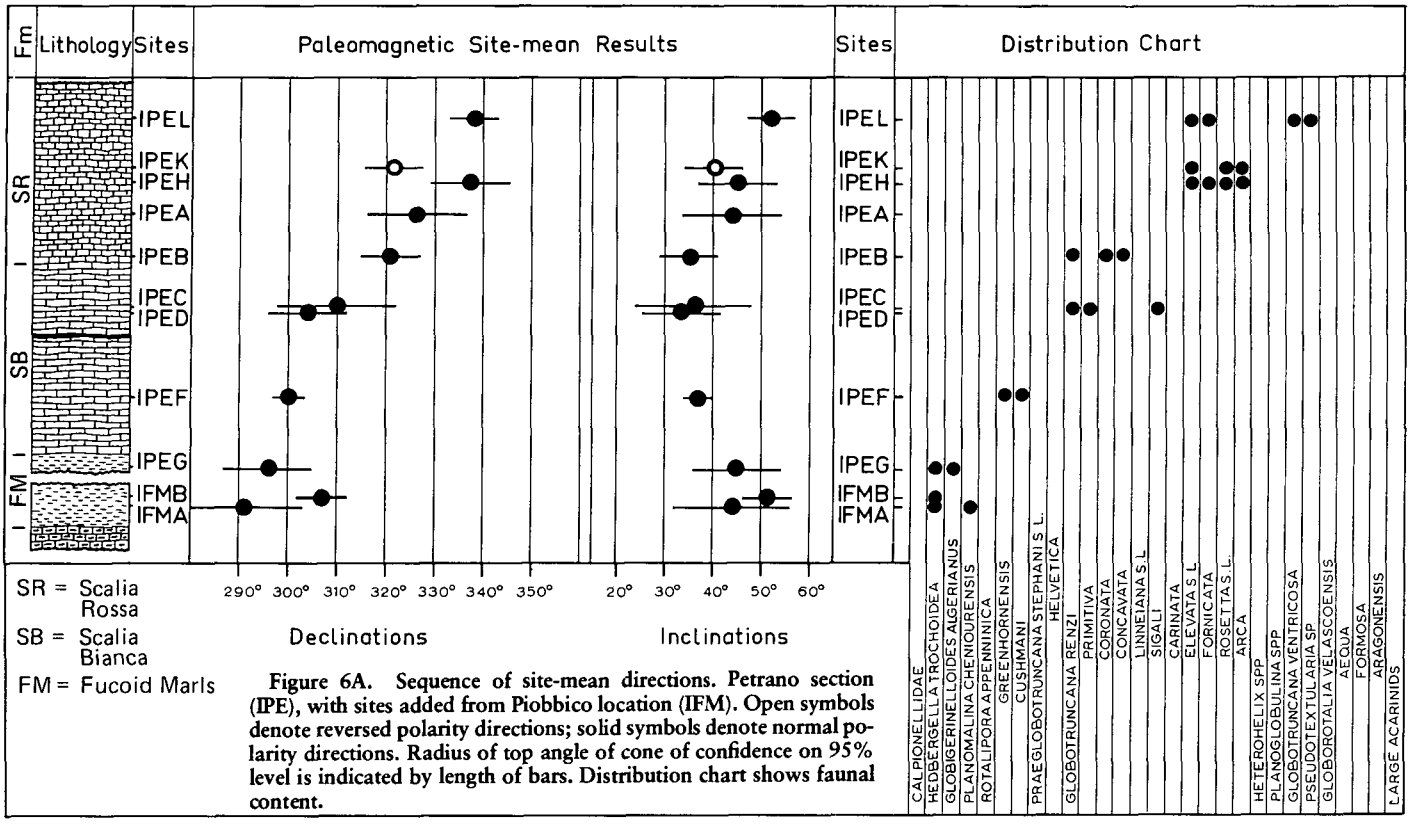


Figure 6A. Sequence of site-mean directions. Petrano section (IPE), with sites added from Piobbico location (IFM). Open symbols denote reversed polarity directions; solid symbols denote normal polarity directions. Radius of top angle of cone of confidence on 95% level is indicated by length of bars. Distribution chart shows faunal content.

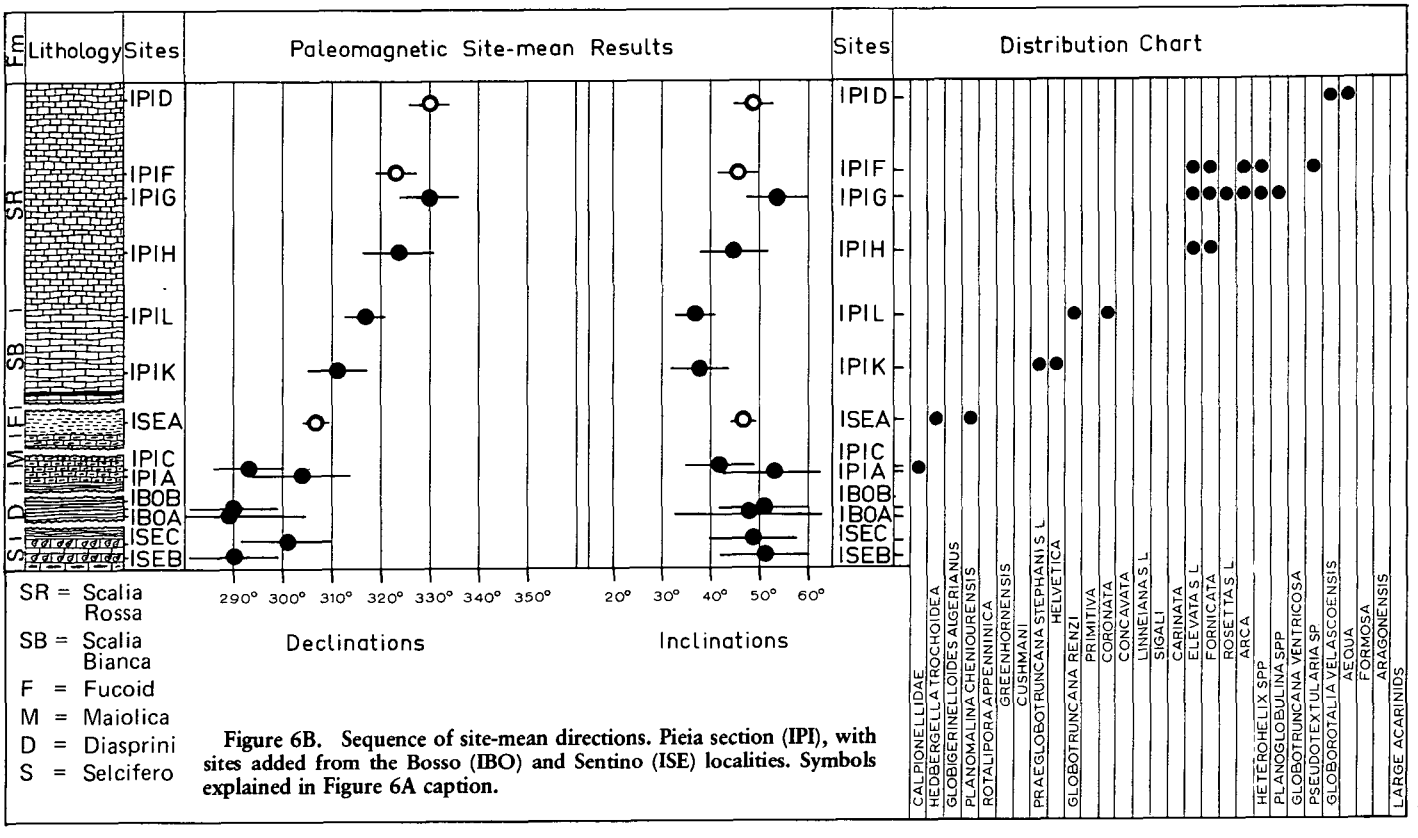


Figure 6B. Sequence of site-mean directions. Pieia section (IPI), with sites added from the Bosso (IBO) and Sentino (ISE) localities. Symbols explained in Figure 6A caption.

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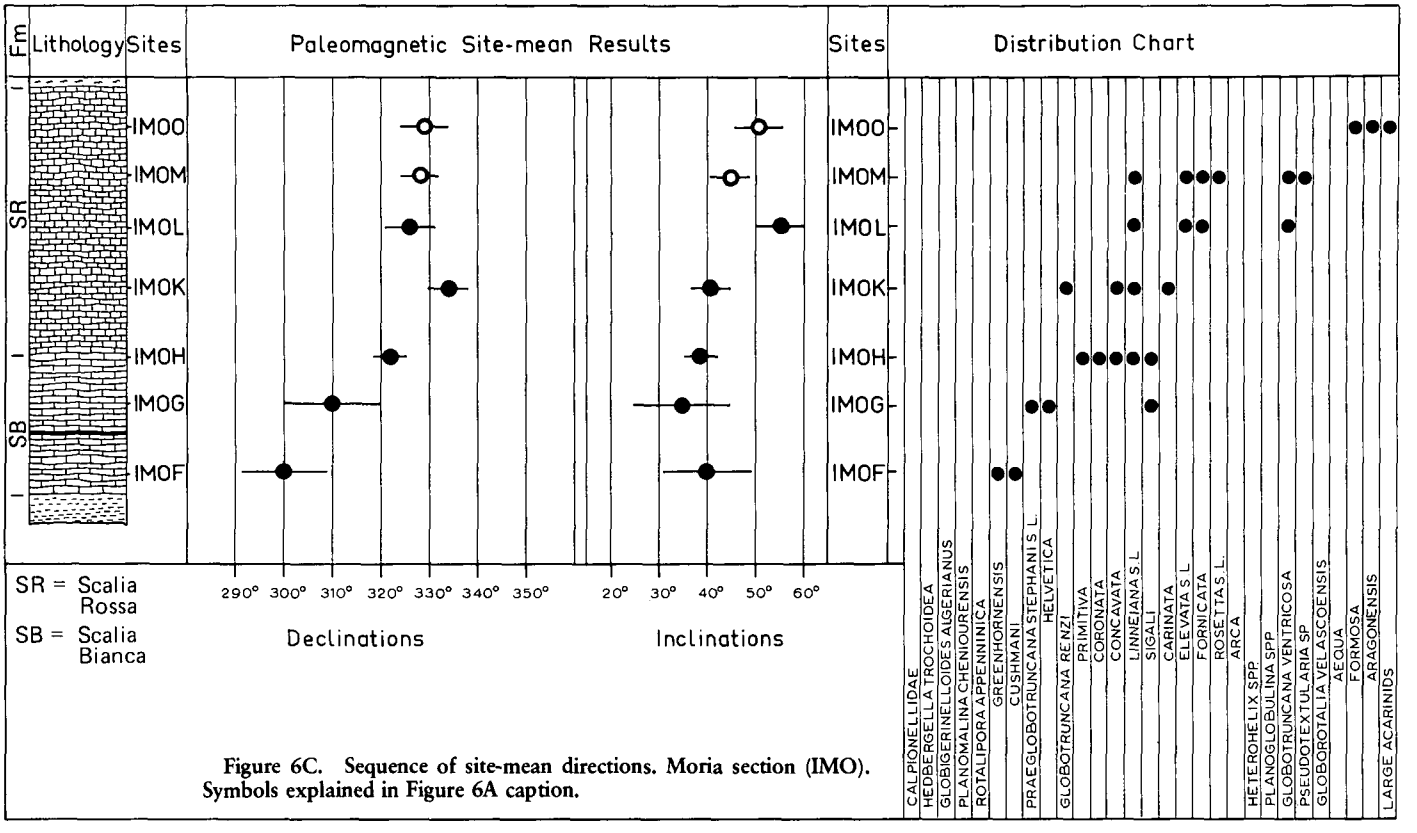


Figure 6C. Sequence of site-mean directions. Moria section (IMO). Symbols explained in Figure 6A caption.

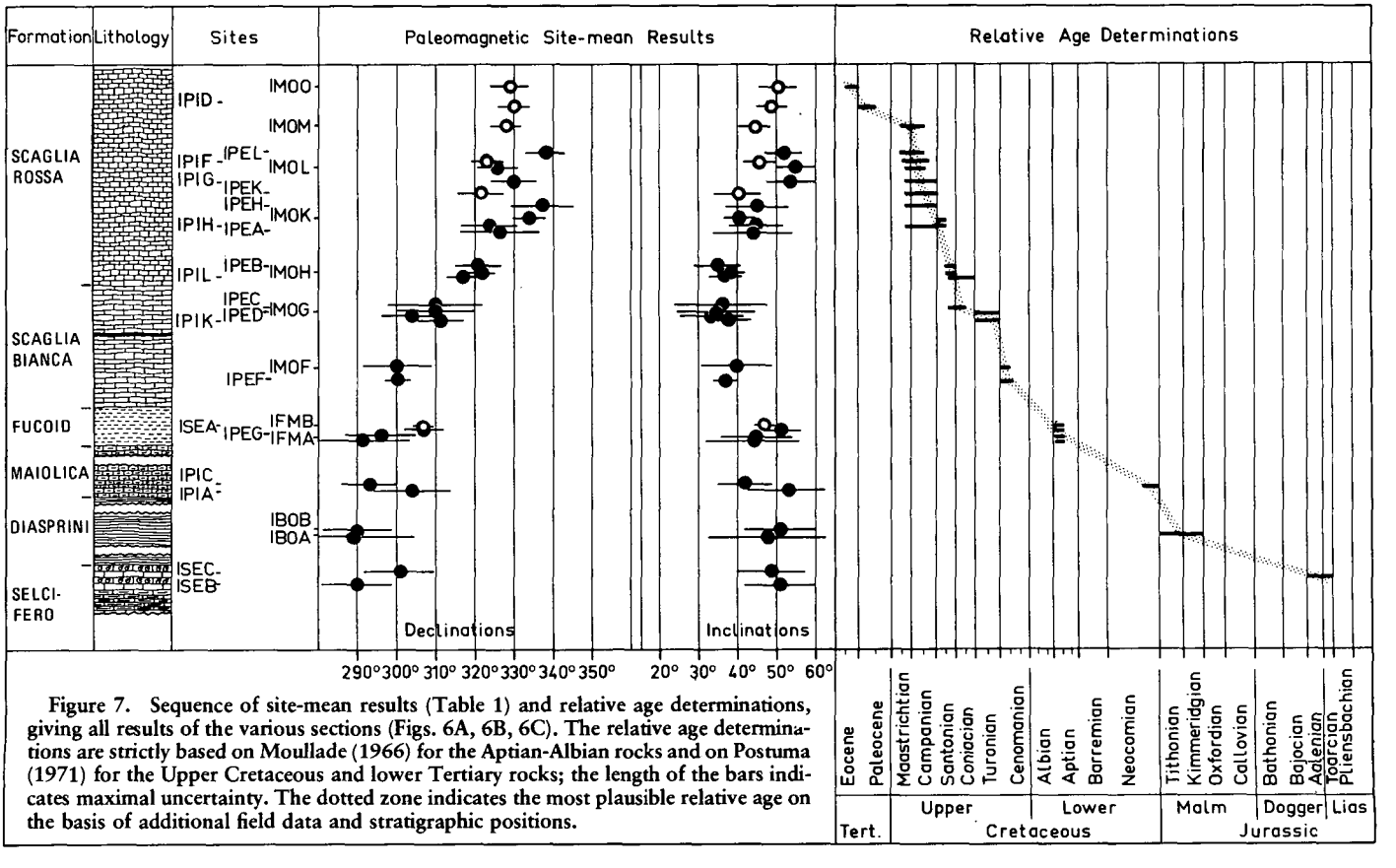


Figure 7. Sequence of site-mean results (Table 1) and relative age determinations, giving all results of the various sections (Figs. 6A, 6B, 6C). The relative age determinations are strictly based on Moullade (1966) for the Aptian-Albian rocks and on Postuma (1971) for the Upper Cretaceous and lower Tertiary rocks; the length of the bars indicates maximal uncertainty. The dotted zone indicates the most plausible relative age on the basis of additional field data and stratigraphic positions.

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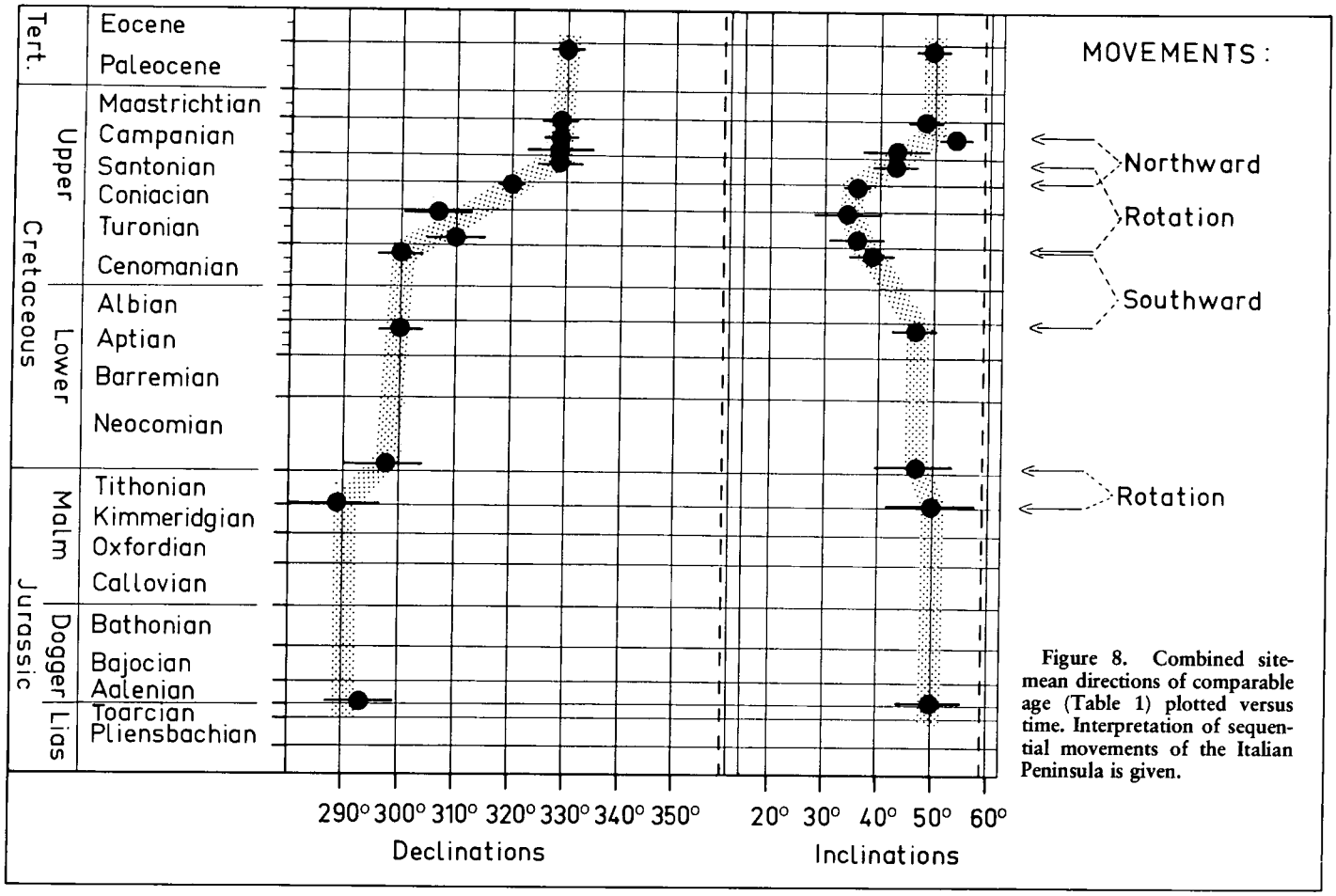


Figure 8. Combined site-mean directions of comparable age (Table 1) plotted versus time. Interpretation of sequential movements of the Italian Peninsula is given.

Globorotalia aequa **Cushman and Renz**

Globorotalia crassata var. *aequa* Cushman and Renz (1942, p. 12, Pl. 3a–c).

Strongly ventroconvex forms with slightly inflated, little ornamented chambers and a narrow umbilicus.

Globorotalia formosa **Bolli**

Globorotalia formosa formosa Bolli (1957, p. 76, Pl. 18, figs. 1–3).

This species differs from *G. velascoensis* by the narrower umbilicus and the absence of strong periumbilical ridges.

Globorotalia aragonensis **Nutall**

Globorotalia aragonensis Nutall (1930, p. 288, Pl. 24, figs. 6–8, 10–11).

In thin sections, G. aragonensis appears very similar to *G. formosa,* from which it differs by a still narrower umbilicus.

Large Acarinids

These strongly ornamented, but nonkeeled forms were found in sample IMOO. They could not be determined at the specific level.

PALEOMAGNETIC RESULTS IN STRATIGRAPHIC ORDER

The paleomagnetic site-mean directions as obtained from the three sections and additional sites, combined with their maximum age ranges and foraminiferal content are shown in Figures 6 and 7. The Petrano section (Fig. 6A), the Pieia section (Fig. 6B), and, less evidently, the Moria section (Fig. 6C) show gradual changes in the directional pattern through the sections.

In our view it would be incorrect to draw conclusions from minor variations in the changes observed in each section. By combining all the data into one master section (Fig. 7), we obtain the best time-averaging of the data. Figure 7, therefore, represents our view of the best overall interpretation of the results displayed stratigraphically.

The main changes in paleomagnetic directions occurred in the Fucoid Marls and the Scaglia Formation. Fortunately, this is the interval with the best-known and most useful planktonic microfaunas, which thus allows accurate dating of the recorded changes.

Figure 7 clearly shows that from at least the Toarcian-Aalenian (upper Lower Jurassic) level — that is, the lower limit of our paleomagnetic coverage — up to the upper Aptian level, no major change in declination and inclination occurred. A minor phase of movement at about Tithonian time may be interpreted from the data shown in Figures 7 and 8. An about 10° clockwise rotation follows from a declination increase from about 290° to about 300°. Considering the inclination curve, it is clear that between the upper Aptian and upper Cenomanian levels, the inclination values decrease by 15° to a 35° value. This mean inclination value characterizes the upper Cenomanian, Turonian, and Coniacian rocks. In the Santonian and Campanian rocks, the mean inclination increases again to 50°. This 50° mean inclination remains characteristic for the interval from the Maastrichtian to at least the lower Eocene strata, that is, the present upper limit of our paleomagnetic record, though the inclination values of individual sites are slightly more dispersed than recorded in the lower part of the sections.

The declination curve, which shows a nearly constant mean value of 300° up to the Turonian level, reveals a gradual increase to 330° in the Turonian, Coniacian, and Santonian rocks. This 330° declination value remains characteristic up to at least the lower Eocene level. It must be noted that our youngest (early Eocene) paleomagnetic data from the Umbrian sequence (D = 330°, I =

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+49.5°, Table 1) are still appreciably different from the present-day magnetic field direction in Umbria (D = 359°, I = +59°).

REVERSAL STRATIGRAPHY

As is also indicated in Figure 7, normal polarity directions prevail throughout the sequence. The few reversed polarities found are of particular interest. The late Aptian reversed direction in the basal Fucoid Marls might be related to the Hissar reversed event as given by McElhinny and Burek (1971) or even to the updated Akoh interval (92 to 104 m.y. B.P., as given by Keating and others, 1975a). Alternatively, this reversed direction might represent the last reversed event before the middle to Late Cretaceous normal polarity interval (McElhinny and Burek, 1971; Irving and Couillard, 1973). This interpretation implies a late Aptian age for the top of the M-sequence of magnetic anomalies, that is, the Keathley-Hawaiian lineations. Such an age is in good agreement with the magnetic anomaly time scale as proposed by Larson and Pitman (1972, 1975) and that of Larson and Hilde (1975). It is hoped that a planned study of the underlying Maiolica beds will clarify these questions about the Cretaceous time scales as pointed out by Baldwin and others (1974).

Other reversed polarity directions were found in sites of Campanian and younger age (Fig. 7). This is in good agreement with the reversed events found by Lowrie and Alvarez (1975) in the Bottaccione gorge section, which were dated by them as late Campanian and late Maastrichtian. The reversal time scale of Premoli Silva and others (1974), based on preliminary data, shows mixed polarities in rocks from the Bottaccione section of Maastrichtian and Paleocene age, but unfortunately gives no clear information on Campanian rocks. The agreement of our Campanian and younger reversed events with Larson and Pitman's (1972, 1975) marine magnetic anomaly time scale as well as with Watkins's (1975) correlation between stratigraphic zones and magnetic polarities is remarkable. For these reasons, and according to our present data, we favor the view that the Cretaceous normal polarity interval began in late Aptian time and ended in Campanian time. This working hypothesis is in conflict with the suggestion of Keating and others (1975a), backed by Sclater and Fischer (1974), that the Cretaceous quiet interval did not end before the beginning of Maastrichtian time. Consequently, they (Keating and others) preferred to interpret their recent findings of reversed polarity directions of Campanian and Aptian age in some Deep Sea Drilling Project cores (Keating and others, 1975b), as particular reversed events within the Late Cretaceous normal polarity interval. Anomalies 33 and 34 of Santonian and possibly Coniacian age, according to Larson and Pitman (1972, 1975), were not detected in the present study, nor were they detected by Lowrie and Alvarez (1975) in the Bottacionne gorge.

COMPARISON WITH RELATED STUDIES IN UMBRIA

Several groups are at present actively involved in research on the Scaglia Formation of the Umbrian sequence, and a number of studies have already been published (Premoli Silva and others, 1974; Lowrie and Alvarez, 1974, 1975; Channell and Tarling, 1975; Klootwijk and VandenBerg, 1975; VandenBerg and others, 1975). Sampling of geographically distributed sites as done by Channell and Tarling (1975) and Lowrie and Alvarez (1975) suffered from the drawback that relative ages were not obtained, since paleontologic data were either not or not rigorously applied. Only the well-known Scaglia section in the Bottaccione gorge near Gubbio was sampled in more detail by Premoli Silva and others (1974) and by Lowrie and Alvarez (1975). The paleomagnetic results of Premoli Silva and others (1974), although well dated, are widely dispersed and admittedly of questionable reliability (as stated by Alvarez and Lowrie, 1975, p. 1587). On the other hand, Lowrie and Alvarez' (1975) data seem much more refined, but as they noticed, the ages assigned to the successive sampling sites might be less accurate. These ages are based on simple linear interpolation through the various biostratigraphic zones distinguished by Renz (1936), and again as they noted, Renz had not been able to separate the three crucial stages Turonian, Coniacian, and Santonian.

In contrast, our sampling was confined to the northernmost outcrop of the Umbrian sequence. Three fresh and well-correlated sections were sampled in detail from road cuts, and the age of each site was established by planktonic foraminiferal analysis as described above.

The magnetization was known to be of a complex nature (Klootwijk and VandenBerg, 1975). Therefore, we separated and interpreted the various magnetic components by means of elaborate progressive AF and thermal demagnetization techniques. The orthogonal projection technique was used; this enabled us to distinguish different components of the natural remanent magnetization in complex multicomponent cases (Zijderveld, 1967, 1975) instead of the less-preferable optimum dispersion method, which was used by Premoli Silva and others (1974), Channell and Tarling (1975), and Lowrie and Alvarez (1974, 1975).

From the clearly visible pattern of gradual changes in declination and inclination (Figs. 7, 8), it is evident that an overall grouping of the Scaglia Rossa data leads to a loss of information despite claims to the contrary (Premoli Silva and others, 1974; Channell and Tarling, 1975). A stratigraphic differentiation of paleomagnetic results was made only for the Bottaccione gorge section (Lowrie and Alvarez, 1975). In contrast to the rather complex pattern that we observed in the data from our Scaglia sections, Lowrie and Alvarez (1975) discerned in their Bottaccionne gorge section a gradual change to higher declination values only. This change was estimated by them to be of Campanian-Maastrichtian age, in contrast to our present Turonian-Santonian dating that was obtained through micropaleontologic study. Close inspection of the site-mean results of Lowrie and Alvarez (1975) and Channell and Tarling (1975) reveals that the pattern of successive movements as found in our data seems to be present in their original data as well.

INTERPRETATION

Mean directions (Table 1, Fig. 8) were computed for the successive intervals that are characterized, as described before, by rather stable mean declination and (or) inclination values. Using the geocentric coaxial dipole formula, successive positions of the sampling area were computed from these data with absolute longitudinal positions, of course, remaining undeterminable; movements that are entirely longitudinal, as documented by Pitman and Talwani (1972), for example, cannot be deduced from the present data.

After extrapolating our Umbrian paleomagnetic data to the whole Italian Peninsula, successive positions as shown in Figure 9 could be computed. Figures 8 and 9 reveal the following sequence of rotational and latitudinal movements.

From at least Toarcian-Aalenian to late Aptian time, the Umbrian depositional basin remained at about lat 30°N, and was rotated about 60° in dextral sense compared with its present orientation (Fig. 9A). A slight movement phase in about Tithonian time seems reflected in an increase in declination of about 10°. This paleomagnetically deduced phase is in accord with geologic knowledge, since Sander (1970, p. 110) noted diastrophic events in the Mediterranean realm at the beginning and at the end of Tithonian time (Figs. 9A, 9B). Moreover, Pitman and Talwani (1972) indicated a distinct change in the movement pattern of the Atlantic-bordering continents by that time. Between late Aptian and late Cenomanian time southward movement of about 10° occurred (Fig. 9C), followed by a 30° counterclockwise rotation that took place during Turonian, Coniacian, and Santonian time (Figs. 9D through F). Beginning in the later stage of this rotation and lasting until a somewhat later date (that is, during Santonian and Campanian time), a 10° northward movement took place (Figs. 9F-9G).

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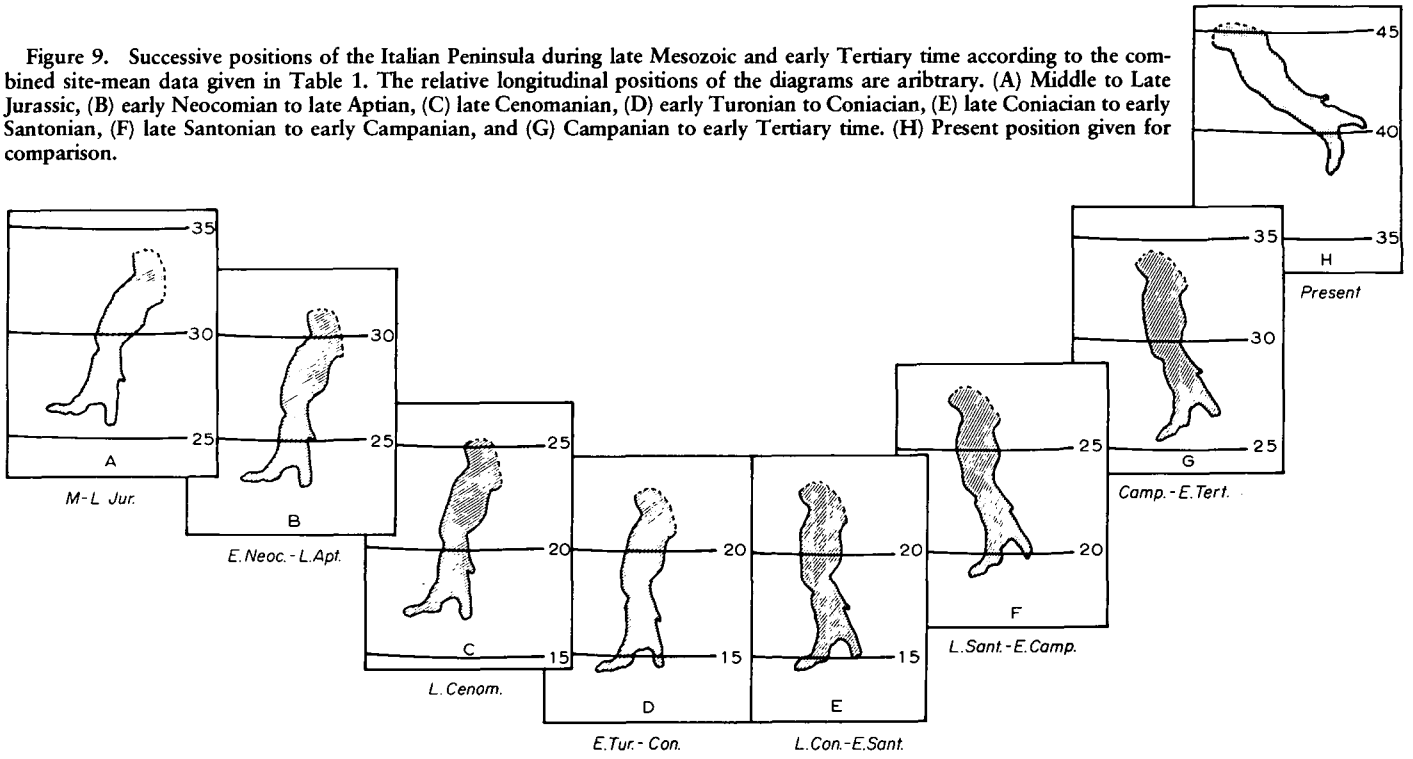


Figure 9. Successive positions of the Italian Peninsula during late Mesozoic and early Tertiary time according to the combined site-mean data given in Table 1. The relative longitudinal positions of the diagrams are aribtrary. (A) Middle to Late Jurassic, (B) early Neocomian to late Aptian, (C) late Cenomanian, (D) early Turonian to Coniacian, (E) late Coniacian to early Santonian, (F) late Santonian to early Campanian, and (G) Campanian to early Tertiary time. (H) Present position given for comparison.

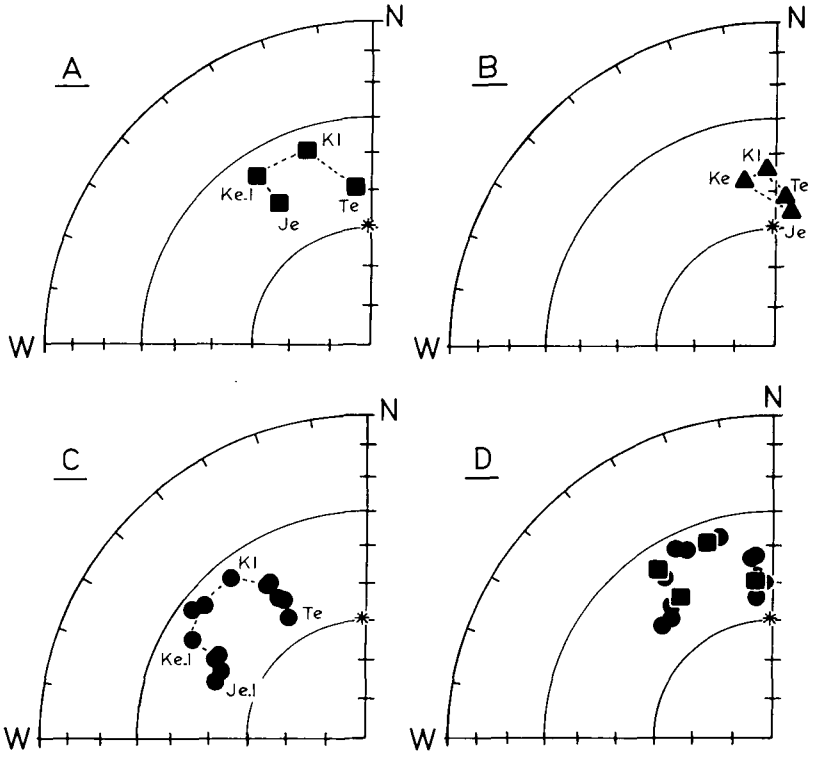


Figure 10. In projection A the northern Apennines are considered as fixed to the African plate. The directions to be expected were computed from the poles for Africa of Van der Voo and French (1974). In projection B the Northern Apennines are considered as fixed to the European plate and the directions to be expected were computed from the poles for Europe of Van der Voo and French (1974). In projection C the present mean directions from the Umbrian sequence are shown (Table 1). Projection D shows the loop formed by the mean directions of the Umbrian sequence (dots) compared with the "African" pattern (squares) after accounting for a post–early Eocene counterclockwise rotation of Italy over about 25° as Italy separated from Africa. Equal-area lower-hemisphere projections. The asterisk denotes the present local field direction. Je-l = Early to Late Jurassic; Ke-l = Early to Late Cretaceous; Te = early Tertiary.

From Maastrichtian to at least late early Eocene time, no marked rotational or meridional movements occurred. Thus, the ultimate northward movement of more than 10° and counterclockwise rotation of about 25° toward the present position (Fig. 9H) are post–early Eocene.

These indications for large-scale tectonic events lead to the interesting question, To what extent is the present paleomagnetic pattern from the Umbrian sequence reflected in the paleomagnetic data from the African and European plates? Unfortunately, a direct comparison of results is hampered by a notable scarcity of Jurassic and Cretaceous data from Europe and Africa. We have tried to overcome this drawback by relying on indirectly determined paleomagnetic poles for the African and European plates. These simulated poles were computed by Van der Voo and French (1974), based on selected paleomagnetic data from the continents bordering the Atlantic, combined with reconstructions of successive stages in the opening of the Atlantic as deduced from sea-floor spreading data by Pitman and Talwani (1972).

The Early Jurassic to early Tertiary succession of paleomagnetic directions from the Umbrian sequence as actually found by us is shown in Figure 10A as a polar equal-area projection of the mean data compiled in Figure 8 and Table 1.

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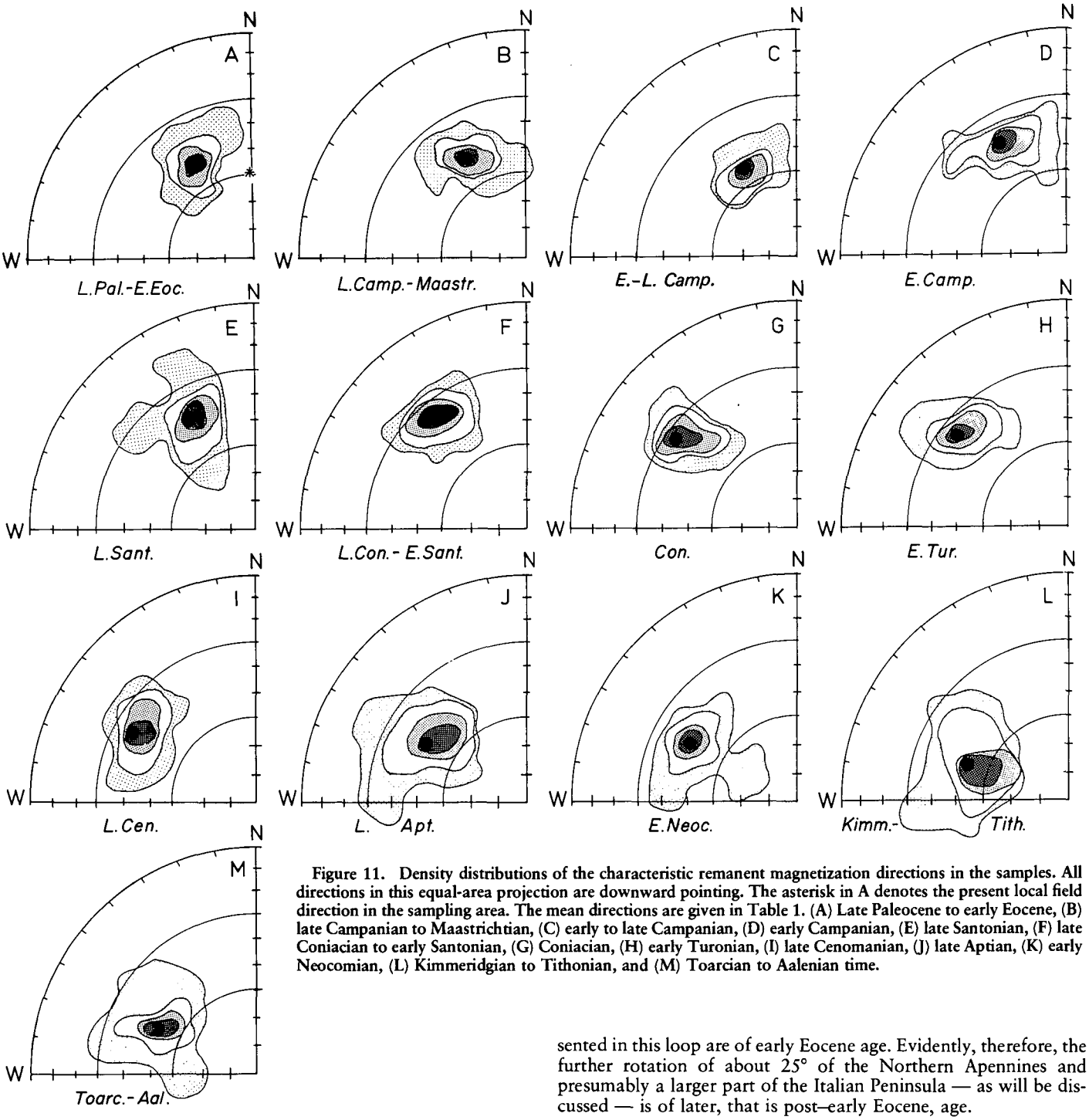


Figure 11. Density distributions of the characteristic remanent magnetization directions in the samples. All directions in this equal-area projection are downward pointing. The asterisk in A denotes the present local field direction in the sampling area. The mean directions are given in Table 1. (A) Late Paleocene to early Eocene, (B) late Campanian to Maastrichtian, (C) early to late Campanian, (D) early Campanian, (E) late Santonian, (F) late Coniacian to early Santonian, (G) Coniacian, (H) early Turonian, (I) late Cenomanian, (J) late Aptian, (K) early Neocomian, (L) Kimmeridgian to Tithonian, and (M) Toarcian to Aalenian time.

For further elucidation, we have represented (Fig. 11) this trend in the form of a series of density distributions of sample directions. In Figures 10A and 10B we depicted the succession of paleomagnetic directions that are to be expected in the Northern Apennines in their present position according to Van der Voo and French's (1974) simulated poles, as extrapolated from Africa and Europe, respectively. The observed Umbrian pattern (Figs. 10C, 11) and the simulated "African" pattern (Fig. 10A) show distinct loops that are of comparable shape and of the same age as well; these patterns contrast with the simulated "European" pattern (Fig. 10B). Despite the remarkable overall conformity between the Umbrian and the "African" pattern, it appears that the actual positions of the loops fit well only after accounting for a further counterclockwise rotation of about 25° of the Umbrian pattern relative to the "African" pattern as shown in Figure 10D. The youngest Umbrian data represented in this loop are of early Eocene age. Evidently, therefore, the further rotation of about 25° of the Northern Apennines and presumably a larger part of the Italian Peninsula — as will be discussed — is of later, that is post–early Eocene, age.

DISCUSSION

It has been shown that Umbria was coupled with the African plate from at least late Early Jurassic until early Eocene time, but that a post–early Eocene 25° counterclockwise rotation of this part of Italy occurred relative to the African plate.

A possible explanation for this separate movement in terms of a supracrustal rotation of the Umbrian sequence, with Triassic evaporite horizons acting as a major décollement surface, does not seem warranted by geologic data. As noted by Lowrie and Alvarez (1975), such a hypothesis would imply improbable horizontal movements in the more southern area of the Umbrian realm. Therefore, basement-inclusive movement is apparently the preferable interpretation, especially because paleomagnetic data from basement rocks in Tuscany support this (VandenBerg and Wonders, 1976). If

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the different magnitudes of rotations documented for graphically distinct areas (Soffel, 1972, 1975a; Channell and Tarling, 1975; VandenBerg and Wonders, 1976) are considered, it is most interesting to trace the extension of the (micro-) plates that underwent a Tertiary counterclockwise rotation relative to Africa.

Concerning the Italian data, clear distinction has to be made between the paleomagnetic data from the Southern Alps and those from areas to the south of the Po plain. This is considered as support for Mueller and Talwani's (1971) suggestion that a profound discontinuity extending to the base of the crust exists between the Apennines and the Southern Alps in the Po plain. Such a zone is also shown by Ritsema (1969).

Delineation of the southern limit of the Italian tectonic unit(s) that rotated during Tertiary time is hampered by lack of data. Late Cretaceous paleomagnetic results from the Capo Passero basalts in southeast Sicily (Schult, 1973; Barberi and others, 1974), convincingly show that no significant post-Cretaceous movements occurred between southeast Sicily and Africa. The only other paleomagnetic data from central and southern Italy are some results from bauxites on the Apulian and Campanian platform (Channell and Tarling, 1975) (see Fig. 12). These data, however, suffer from the drawback common to bauxite deposits that their age is only known within wide limits. The presence of a reversed polarity in the data of Campania and the appreciably younger dating of the bauxites in Campania-Lucania (Grandjacquet, 1963, p. 188–189) point to a Santonian-Campanian age, rather than to the suspected early Late Cretaceous age. The controversial ages give rise to some ambiguity. The paleomagnetic data for the bauxites, if they are of early Late Cretaceous age, are inconclusive as to whether a Tertiary rotation for this part of Italy may be expected. The most probable Late Cretaceous age, on the other hand, undoubtedly indicates the paleomagnetic data of the bauxites to accord with the Umbrian pattern, in which case, a Tertiary counterclockwise rotation for this area as well can be concluded. Manzoni (1975) reported paleomagnetic data (Fig. 12) from Northern Calabria that indicate such a rotation, but these data are not conclusive either, since they were taken from rocks in a nappe structure. If this ambiguity is considered, we cannot exclude the possibility that the Anzio-Ancona line acted as the southern limit of the rotated unit. Nardi's (1971) interpretation of this line as a major thrust fault, however, seems a more likely explanation on the grounds of the present data. In the absence of more limiting paleomagnetic data, we suggest that the southern boundary of the rotated unit has to be searched for farther south, most probably in the South Calabrian arc where the regional geology points to a transverse tectonic zone (Alvarez and others, 1974). Alternatively, it may be still slightly farther south on Sicily's northern margin (Barberi and others, 1974).

As to the western extension of the Italian plate, it can be noted that all Sardinian data fit well on the apparent polar wander path for the Italian Peninsula, whereas the Corsican data do not. This opens the interesting perspective that the late Oligocene–early Miocene or even earlier rotation of Sardinia (DeJong and others, 1973; Coulon and others, 1974) is related to the post–early Eocene rotation of the Italian Peninsula (VandenBerg and Wonders, 1976).

To the east, the Dinarides, Hellenides, and Central European and Balkan Alpine orogenic belts constitute a complex pattern of possible plate boundaries. Since very few, if any, paleomagnetic data from these areas are available, the question about the eastern boundary of the Italian Peninsula microplate is open to further research.

On the basis of paleomagnetic data available at present, it can tentatively be concluded that during late Mesozoic times, the Italian Peninsula rotated and moved together with and as part of the African plate. Although the movements in the various parts of the Italian border zone of the African plate were mutually coherent, in detail, a certain individuality in movements seems apparent. Another subject of further research, therefore, is to what extent this absence of rigidity warrants the distinction and delineation of separate microplates.

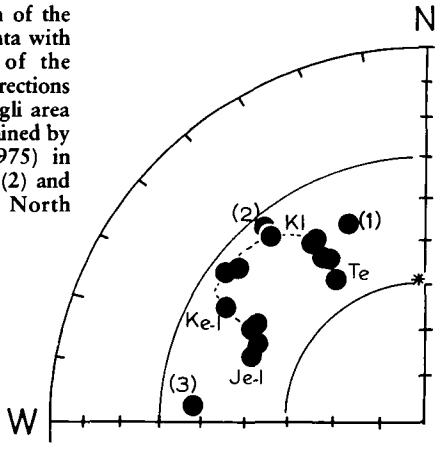


Figure 12. Comparison of the Umbrian paleomagnetic data with data from areas south of the Anzio-Ancona line. The directions were computed for the Cagli area from the virtual poles obtained by Channell and Tarling (1975) in Apulia (1) and Campania (2) and by Manzoni (1975) in North Calabria (3).

MEDITERRANEAN AND ATLANTIC RELATIONSHIPS

The good agreement of the conspicuous loops of the simulated "African" paleomagnetic pattern for Umbria and the actually observed Umbrian pattern clearly indicates a direct relationship between the spreading history of the North Atlantic (Le Pichon and Fox, 1971; Smith, 1971; Phillips and Forsyth, 1972; Pitman and Talwani, 1972; Larson and Pitman, 1972, 1975; Dewey and others, 1973) and the proposed sequence of movements of the Italian Peninsula (Figs. 10A, 10C, 10D). Pitman and Talwani (1972) and Dewey and others (1973) recognized three main phases of movement between Europe and Africa: (1) (180 to 80 m.y. B.P.) Since the Early Jurassic beginning of the opening of the North Atlantic, Africa underwent a southeastward-directed divergent movement relative to Europe. A counterclockwise rotation of Africa started in the second half of this stage. (2) (80 to 53 m.y. B.P.) The relative movement changed with Africa moving westward relative to Europe. Until about 63 m.y. B.P. there was continuing counterclockwise rotation of Africa and compression between Europe and Africa. (3) (53 m.y. B.P. to the present) The movement essentially resulted in compression.

The pattern of sequential movements of the Italian Peninsula as established from our Umbrian paleomagnetic data accords rather well with this scheme, and corresponding movements can be deduced for the Southern Alps (VandenBerg and Wonders, 1976). As far as the present Umbrian paleomagnetic data are concerned, the southeastward movement of Africa (1, above) seems reflected in the minor movement phase in about Tithonian time and in the late Aptian to late Cenomanian component of southward movement of the Italian Peninsula. The first rotation of our two-phase rotational hypothesis, that is, the Turonian to Santonian 30° counterclockwise rotation, is contemporaneous with the rotational movement of the African plate as mentioned above (1 and 2), which supports our conclusion that Italy was part of the African plate during that time. Therefore, we disagree with Lowrie and Alvarez (1974) and do not think that this earlier rotation resulted from microplate adjustment to the predominantly compressive Late Cretaceous–Tertiary motion between Africa and Europe.

Rotation of Italy according to the ball-bearing concept as originally suggested by VanderVoo and Zijderveld (1969), elaborated by Smith (1971), and later on somewhat retracted (Zijderveld and VanderVoo, 1973; Lowrie and Alvarez, 1975) can be ruled out indeed. The movement of Africa relative to Europe was essentially dextral during the Turonian to Santonian rotation of the Italian Peninsula. According to the ball-bearing concept, however, one would expect a rotation in the opposite sense.

The paleomagnetically determined Santonian-Campanian northward movement of the Italian Peninsula, followed later on by a post–early Eocene phase of renewed northward movement agrees with the compressive phases mentioned above (2 and 3). In contrast to the first rotational movement phase of Italy as a coherent part of the African plate, the post–early Eocene rotation of Italy was

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presumably caused by microplate adjustment to compressive movement between the northward-moving African plate.

The extensive development of graben systems was described by Le Pichon and others (1971) in Sardinia, along the Mediterranean coast of France, and in more internal European regions as well. According to these authors, such graben formation postdated a paroxysmal phase in the Alpine orogenesis. It is probable, however, that the grabens are in fact related to the second rotation phase of Italy.

Mainly in the eastern Mediterranean, but in the western Mediterranean as well, Argyriadis (1974) could easily discern, in addition to the Tertiary paroxysmal Alpine phase, earlier tectonic phases of middle and Late Cretaceous age. Also Caby (1975) discerned two main phases of tectonic activity from analysis of patterns of transverse and longitudinal folds in the Western Alps. This polyphase model is also recognizable in the metamorphic events of the Alps (Jäger, 1973; Hawkesworth and others, 1974), the Apennines (Dal Piaz, 1974) and even in the Betic Cordilleras (Egeler, 1975).

It would be of interest to test whether other parts of the Mediterranean show an analogous two-phase development. For Spain, for instance, it is well known that available paleomagnetic data are seemingly in agreement with the African apparent polar wander path, but it cannot be established from these data whether Spain rotated as part of the African plate as documented here for peninsular Italy. More convincing in this respect is the recent interpretation of the opening of the Bay of Biscay by Williams (1975). This opening was dated between Barremian and Maastrichtian time, whereas according to Wilson (1975), most of the movement occurred between Aptian and Cenomanian time.

CONCLUSIONS

The dated paleomagnetic record of the Umbrian sequence details polyphase rotational movements of peninsular Italy during late Mesozoic and Cenozoic time. From Toarcian-Aalenian to late Aptian time, the paleomagnetic data do not indicate major rotational and latitudinal movements except for a minor Tithonian counterclockwise rotation of about 10°. During late Aptian to late Cenomanian time, a further southward movement over about 10° of latitude took place. A major phase of counterclockwise rotation over about 30° occurred during Turonian, Coniacian, and Santonian time, followed by a Santonian-Campanian northward movement over about 10° of latitude. From Maastrichtian to at least early Eocene time, no significant rotational and latitudinal movements occurred. The ultimate 10° northward and additional 25° counterclockwise rotation cannot be dated more closely than post–early Eocene.

The conspicuous loops of the actual Umbrian paleomagnetic pattern agree well with a simulated paleomagnetic pattern for Umbria that is based on the spreading history of the North Atlantic and on paleomagnetic data from the Atlantic-bordering continents. This indicates that peninsular Italy moved as part of the African plate during at least late Mesozoic time. The later post–early Eocene rotation of peninsular Italy occurred separately from movement of the African plate and was caused presumably by microplate adjustment to compression between Europe and the northward-moving African plate.

Much more study is needed to delineate the complex pattern of microplate movements in the Alpine-Mediterranean realm. The present data reveal the utility of paleomagnetic studies of dated carbonate sequences for detailing the tectonic history of the Mediterranean area.

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REFERENCES CITED

Alvarez, W., Cocozza, T., and Wezel, F. C., 1974, Fragmentation of the Alpine orogenic belt by microplate dispersal: Nature, v. 248, p. 309–314.

Argyriadis, I., 1974, Sur l'orogenèse Mésogéene des temps crétacés: Rev. Géographie Phys. et Géologie Dynam., v. 16, p. 23–60.

Baldwin, B., Coney, P. J., and Dickinson, W. R., 1974, Dilemma of a Cretaceous time scale and rates of sea-floor spreading: Geology, v. 2, p. 267–270.

Barberi, F., Cesetta, L., Gasparini, P., Innocente, F., Scandone, R., and Villari, L., 1974, Evolution of a section of the Africa-Europe plate boundary; paleomagnetic and volcanological evidence from Silicy: Earth and Planetary Sci. Letters, v. 24, p. 123–132.

Bolli, H. M., 1945, Zur Stratigraphie der oberen Kreide in den höheren Helvetischen Decken: Eclogae Geol. Helvetiae, v. 37, p. 217–326.

——1957, The genera *Praeglobotruncana, Rotalipora, Globotruncana* and *Abathomphalus* in the Upper Cretaceous of Trinidad, B.W.I.: U.S. Natl. Mus. Bull., v. 215, p. 61–82.

Bortolotti, V., Passerini, P., Sagri, M., and Sestini, G., 1970, The miogeosynclinal sequences, *in* Sestini, G., ed. Development of the Northern Apennines geosyncline: Sed. Geology, v. 4, p. 314–444.

Brotzen, F., 1934, Foraminiferen aus dem Senoon Palästinas: Zeitschr. Deutsch. Palästina Ver. v. 57, p. 28–72.

Caby, R., 1975, Geodynamic implications of transverse folding in the Western Alps for the Alpine fold belt: Nature, v. 256, p. 114–117.

Carsey, D. O., 1926, Foraminifera of the Cretaceous of Central Texas: Texas Univ. Bull., no. 2612, p. 1–56.

Channell, J.E.T., and Tarling, D. H., 1975, Paleomagnetism and the rotation of Italy: Earth and Planetary Sci. Letters, v. 25, p. 177–188.

Coulon, C., Demant, A., and Bobier, C., 1974, Contribution du paléomagnetisme à l'étude des séries volcaniques cénozoiques et quarternaires de Sardaigne nord-occidentale. Le problème de la dérive de la Sardaigne: Tectonophysics, v. 22, p. 59–82.

Cushman, J. A., 1925, Some new Foraminifera from the Velasco shale of Mexico: Contr. Cushman Lab. Foram. Research, v. 1, no. 1, p. 18–23.

——1926, Some Foraminifera from the Mendez shale of eastern Mexico: Contr. Cushman Lab. Foram. Research, v. 2, pt. 1, p. 16–26.

Cushman, J. A., and Renz, H. H., 1942, Eocene, Midway, Foraminifera from Soldado Rock, Trinidad: Contr. Cushman Lab. Foram. Research, v. 18, p. 1–22.

Cushman, J. A., and Ten Dam, A., 1948, *Globigerinelloides,* a new genus of the Globigerinidae: Contr. Cushman Lab. Foram. Research, v. 24, pt. 2, p. 42–43.

Dalbiez, F., 1955, The genus *Globotruncana* in Tunisia: Micropaleontology, v. 1, p. 161–171.

Dal Piaz, G. V., 1974, Le métamorphisme de haute pression et basse température dans l'évolution structurale du bassin ophiolitique alpino-apenninique (2me partie): Schweizer. Mineralog. u. Petrog. Mitt., v. 54, p. 399–424.

Dewey, J. F., Pitman, W. C., III, Ryan, W.B.F., and Bonnin, J., 1973, Plate tectonics and the evolution of the Alpine systems: Geol. Soc. America Bull., v. 84, p. 3137–3180.

Egeler, C. G., 1975, Alpine metamorphism and orogenic evolution, *in* Borradaile, G. J., Ritsema, A. R., Rondeel, H. E., and Simon, O. J., eds., Progress in Geodynamics: Amsterdam, Royal Netherlands Acad. Arts and Sciences, Geodynamics Proj. Sci. Rept. no. 13, p. 85–89.

Fazzini, P., 1973, Pieghe minori nella "Scaglia" Umbro-Marchigiana: Soc. Geol. Italiana Boll., v. 92, p. 473–483.

Gandolfi, R., 1942, Ricerche micropaleontologiche e stratigrafiche sulla scaglia e sul flysch cretacici dei dintorni di Balerna (canton Ticino): Riv. Italiana Paleontologia, v. 48, mem. 4, p. 1–160.

Grandjacquet, C., 1963, Schema structural de l'Apennin Campano-Lucanien (Italie): Rev. Géographie Phys. et Géologie Dynam., ser. 2, v. 5, pt. 3, p. 185–202.

Hawkesworth, C. J., Waters, D. J., and Bickle, M. J., 1975, Plate tectonics in the Eastern Alps: Earth and Planetary Sci. Letters, v. 24, p. 405–413.

Irving, E., and Couillard, R. W., 1973, Cretaceous normal polarity interval: Nature Phys. Sci., v. 244, p. 10–11.

Jäger, E., 1973, Die Alpine Orogenese im Lichte der radiometrischen Alterbestimmung: Eclogae Geol. Helvetiae, v. 66, p. 11–21.

Jelinek, V., 1966, A high sensitivity spinner magnetometer: Studia Geophysica et Geodetica, v. 10, p. 58–78.

de Jong, K. A., Manzoni, M., Stavenga, T., van Dijk, F., Van der Voo, R., and Zijderveld, J.D.A., 1973, Paleomagnetic evidence for the rotation of Sardinia during the early Miocene: Nature, v. 243, p. 281–283.

Downloaded from gsabulletin.gsapubs.org on October 21, 2010

Keating, B., Helsley, C. E., and Pessagno, E. A., Jr., 1975a, Late Cretaceous reversal sequence: Geology, v. 3, p. 73–76.
——1975b, Reversed events within the Late Cretaceous normal polarity interval [abs.]: EOS (Am. Geophys. Union Trans.), v. 56, p. 364.
Klootwijk, C. T., and VandenBerg, J., 1975, The rotation of Italy: Preliminary paleomagnetic data from the Umbrian sequence, Northern Apennines: Earth and Planetary Sci. Letters, v. 25, p. 263–273.
Larson, R. L., and Hilde, T.W.C., 1975, A revised time scale of magnetic reversals for the Early Cretaceous and Late Jurassic: Jour. Geophys. Research, v. 80, p. 2586–2594.
Larson, R. L., and Pitman, W. C., III, 1972, World-wide correlation of Mesozoic magnetic anomalies, and its implications: Geol. Soc. America Bull., v. 83, p. 3645–3662.
——1975, World-wide correlation of Mesozoic magnetic anomalies and its implications: Discussion and reply: Geol. Soc. America Bull., v. 86, p. 270–272.
Lowrie, W., and Alvarez, W., 1974, Rotation of the Italian Peninsula: Nature, v. 251, p. 285–288.
——1975, Paleomagnetic evidence for the rotation of the Italian Peninsula: Jour. Geophys. Research, v. 80, p. 1579–1592.
Luterbacher, H. P., 1964, Studies on some *Globorotalia* from the Paleocene and lower Eocene of the Central Apennines: Eclogae Geol. Helvetiae, v. 57, p. 631–730.
Luterbacher, H. P., and Premoli Silva, I., 1962, Note préliminaire sur une révision du profil du Gubbio, Italie: Riv. Italiana Paleontologia, v. 68, p. 253–288.
——1964, Biostratigrafia del limite Cretaceo-Terziario nell'Apennino Centrale: Riv. Italiana Paleontologia, v. 70, p. 67–128.
Manzoni, M., 1975, Rotation of Calabria: Paleomagnetic evidence: Geophys. Research Letters, v. 2, p. 427–429.
McElhinny, M. W., and Burek, P. J., 1971, Mesozoic palaeomagnetic stratigraphy: Nature, v. 232, p. 98–102.
McKenzie, D. P., 1970, Plate tectonics of the Mediterranean region: Nature, v. 226, p. 239–243.
Morrow, A. L., 1934, Foraminifera and Ostracoda from the Upper Cretaceous of Kansas: Jour. Paleontology, v. 8, p. 186–205.
Moullade, M., 1966, Etude stratigraphique et micropaléontologique du crétacé inférieur de la "fosse vocontienne": Lyons, Doc. Lab. Géologie Fac. Sci., v. 15, p. 369.
Mueller, S. , and Talwani, M., 1971, A crustal section across the Eastern Alps based on gravity and seismic refraction data: Pure and Appl. Geophysics, v. 85, p. 226–239.
Nardi, L. D., Elter, P., and Nardi, R., 1971, Considerazioni sull'arco dell'Apennino e sulla "Linea" Ancona-Anzio: Soc. Geol. Italiana Boll., v. 90, p. 203–211.
Nutall, W. L., 1930, Notes on the Tertiary Foraminifera of southern Mexico: Jour. Paleontology, v. 4, pt. 3, p. 271–293.
d'Orbigny, A., 1839, Foraminifères, *in* de la Sagra, R., ed., Histoire physique, politique et naturelle de l'ile de Cuba: Paris, A. Bertrand, p. 1–224.
Phillips, J. D., and Forsyth, D., 1972, Plate tectonics, paleomagnetism, and the opening of the Atlantic: Geol. Soc. America Bull., v. 83, p. 1579–1600.
Le Pichon, X., and Fox, P. J., 1971, Marginal offsets, fracture zones, and the early opening of the North Atlantic: Jour. Geophys. Research, v. 76, p. 6294–6308.
Le Pichon, X., Pautot, G., Auzende, J. M., and Olivet, J. L., 1971, La Méditerranée occidentale depuis l'Ogligoène; Schéma d'évolution: Earth and Planetary Sci. Letters, v. 13, p. 145–152.
Pitman, W. C., III, and Talwani, M., 1972, Sea-floor spreading in the North Atlantic: Geol. Soc. America Bull., v. 83, p. 619–646.
Plummer, H. J., 1931, Some Cretaceous Foraminifera in Texas: Texas Univ. Bull., no. 3101, p. 109–203.
Postuma, J., 1971, Manual of planktonic Foraminifera: Amsterdam Elsevier Pub. Co.
Premoli Silva, I., Napoleone, G., and Fischer, A. G., 1974, Risultate preliminare sulla stratigrafia paleomagnetica della Scaglia Cretaceo-Paleocenica della sezione di Gubbio (Apennino Centrale): Soc. Geol. Italiana Boll., v. 93, p. 647–659.
Reichel, M., 1950, Observations sure les *Globotruncana* du gisement de la Breggia: Eclogae Geol. Helvetiae, v. 42, p. 596–617.
Renz, O., 1936, Stratigraphische und mikropaläontologische Untersuchung der Scaglia (obere Kreide-Tertiär) im Zentralen Apennin: Eclogae Geol. Helvetiae, v. 29, p. 1–149.
Ritsema, A. R., 1969, Seismic data of the West Mediterranean and the problem of oceanization, *in* Symposium on the problem of oceanization of the Western Mediterranean: Verh. Koninkl Nederlands Geologieen Mijnbouw Genoot., v. 26, p. 105–120.
Sander, N. J., 1970, Structural evolution of the Mediterranean region during the Mesozoic era, *in* Alvarez, W., and Gohrbrandt, K.M.A., eds., Geology and history of Sicily: Tripoli, The Petroleum Exploration Society of Libya, p. 43–132.
Schult, A., 1973, Paleomagnetism of Upper Cretaceous volcanic rocks in Sicily: Earth and Planetary Sci. Letters, v. 19, p. 97–100.
Sclater, J. G., and Fischer, R. L., 1974, Evolution of the east central Indian Ocean, with emphasis on the tectonic setting of the Ninetyeast Ridge: Geol. Soc. America Bull., v. 85, p. 683–702.
Sigal, J., 1966, Le concept taxonomique de spectre: Soc. Géol. France Mém. hors-série 3, p. 1–126.
Smith, A. G., 1971, Alpine deformation and the oceanic areas of the Tethys, Mediterranean and Atlantic: Geol. Soc. America Bull., v. 82, p. 2039–2070.
Soffel, H., 1972, Anticlockwise rotation of Italy between the Eocene and Miocene: Palaeomagnetic evidence from the Colli Euganei, Italy: Earth and Planetary Sci. Letters, v. 17, p. 297–310.
——1974, Paleomagnetism and rock magnetism of the Colli Euganei volcanites and the rotation of Northern Italy between Eocene and Oligocene: Boll. Geofisica Teor. ed Appl., v. 16, p. 333–355.
——1975a, The palaeomagnetism of age dated Tertiary volcanites of the Monte Lessini (Northern Italy) and its implication to the rotation of Northern Italy: Jour. Geophys., Germ., v. 41, pt. 4, p. 385–400.
——1975b, Rock magnetism of the Monte Lessini and Monte Berici volcanites and age of volcanism deduced from the Heirtzler polarity time scale: Jour. Geophys., Germ., v. 41, pt. 4, p. 401–411.
VandenBerg, J., and Wonders, A.A.H., 1976, Paleomagnetic evidence of large fault displacements around the Po-basin, *in* Briden, J. C., ed., Ancient plate margins: Tectonophysics, v. 33, p. 301–320.
VandenBerg, J., Klootwijk, C. T., and Wonders, A.A.H., 1975, Implications for the rotational movement of Italy from current palaeomagnetic research in the Umbrian sequence, Northern Apennines, *in* Borradaile, G. J., Ritsema, A. R., Rondeel, H. E., and Simon, O. J., eds., Progress in Geodynamics: Amsterdam, Royal Netherlands Acad. Arts and Sciences, Geodynamics Proj. Sci. Rept. no. 13, p. 165–175.
Van der Voo, R., and French, R. B., 1974, Apparent polar wandering for the Atlantic-bordering continents: Late Carboniferous to Eocene: Earth Sci. Rev., v. 10, p. 99–119.
Van der Voo, R., and Zijderveld, J.D.A., 1969, Paleomagnetism in the Western Mediterranean area, *in* Ritsema, A. R., ed., Symposium on the problem of oceanization in the Western Mediterranean: Verh. Koninkl. Nederlands Geologie en Mijnouw Genoot, v. 26, p. 121–138.
Watkins, N. D., 1975, Correlating stratigraphic zones and magnetic polarities: Geotimes, June, p. 26–27.
White, M. P., 1928, Some index Foraminifera of the Tampico Embayment area of Mexico: Jour. Paleontology, v. 2, p. 280–317.
Williams, C. A., 1975, Sea-floor spreading in the Bay of Biscay and its relationship to the North Atlantic: Earth and Planetary Sci. Letters, v. 24, p. 440–456.
Wilson, R.C.L., 1975, Atlantic opening and Mesozoic continental margin basins of Iberia: Earth and Planetary Sci. Letters, v. 25, p. 33–43.
Zijderveld, J.D.A., 1967, AC-demagnetization of rocks: Analysis and results, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., eds., Methods in palaeomagnetism: Amsterdam, Elsevier Pub. Co., p. 254–286.
——1975, Paleomagnetism of the Esterel rocks [Ph.D. thesis]: Utrecht, The Netherlands, State Univ. Utrecht, 209 p.
Zijderveld, J.D.A., and Van der Voo, R., 1973, Palaeomagnetism in the Mediterranean area, *in* Tarling, D. H., and Runcorn, S. K., eds., Implications of continental drift to the earth sciences, Vol. 1: London, Academic Press, p. 133–161.

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